

Algebraic Geometry 10

Lecture notes with examples

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Algebraic Geometry 10

Divisor Class Groups, Morphisms of Projective Spaces, Coherent Sheaves,
and Closed Subschemes

Contents

1 Closed Immersions Defined by Linear Systems

Solution. Let k be algebraically closed, let $X \subseteq \mathbb{P}_k^m$ be closed, and let

$$\varphi : X \longrightarrow \mathbb{P}_k^n$$

be the morphism defined by global sections

$$s_0, \dots, s_n \in \Gamma(X, \mathcal{L}),$$

where $\mathcal{L}$ is generated by these sections. Let

$$V = \langle s_0, \dots, s_n \rangle \subseteq \Gamma(X, \mathcal{L}).$$

We prove that φ is a closed immersion if and only if:

(a) V separates points,

and

(b) V separates tangent vectors.

Step 1: If φ is a closed immersion

Assume first that

$$\varphi : X \hookrightarrow \mathbb{P}_k^n$$

is a closed immersion.

A closed immersion is injective on closed points, so if $P \neq Q$, then

$$\varphi(P) \neq \varphi(Q).$$

Since the map is given by

$$P \mapsto [s_0(P) : \dots : s_n(P)],$$

there must exist some section $s_i \in V$ such that

$$s_i(P) \neq s_i(Q)$$

after identifying the fibers through suitable local trivializations. Equivalently, there exists

$$s \in V$$

such that

$$s_P \notin \mathfrak{m}_P \mathcal{L}_P \quad \text{and} \quad s_Q \in \mathfrak{m}_Q \mathcal{L}_Q,$$

or the reverse. Thus V separates points.

Now fix a closed point $P \in X$. Since φ is a closed immersion, it induces an isomorphism on Zariski tangent spaces:

$$T_P X \cong T_{\varphi(P)} \varphi(X).$$

Equivalently, the induced map on cotangent spaces

$$\mathfrak{m}_{\varphi(P)} / \mathfrak{m}_{\varphi(P)}^2 \longrightarrow \mathfrak{m}_P / \mathfrak{m}_P^2$$

is surjective.

The local affine coordinates on $\mathbb{P}_k^n$ are ratios of the sections s_i/s_j . Therefore their first-order variations generate the cotangent space of X at P . This says exactly that

$$\{s \in V \mid s_P \in \mathfrak{m}_P \mathcal{L}_P\}$$

spans

$$\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P.$$

Hence V separates tangent vectors.

Step 2: If V separates points and tangent vectors

Assume now that conditions (a) and (b) hold.

Since X is projective, the morphism

$$\varphi : X \rightarrow \mathbb{P}_k^n$$

is proper. Therefore $\varphi(X)$ is closed in $\mathbb{P}_k^n$.

By condition (a), distinct closed points of X are separated by sections in V . Hence

$$\varphi(P) \neq \varphi(Q)$$

for all distinct closed points $P, Q \in X$. Thus φ is injective on closed points.

It remains to check that the morphism is locally a closed immersion. This is done by checking the induced map of sheaves

$$\varphi^\# : \mathcal{O}_{\mathbb{P}_k^n} \longrightarrow \varphi_* \mathcal{O}_X.$$

On affine charts of $\mathbb{P}_k^n$, the coordinate functions are generated by ratios

$$\frac{s_i}{s_j}.$$

Condition (b) says that these functions generate all first-order directions at every point of X . Equivalently, for every closed point $P \in X$, the induced map on cotangent spaces is surjective:

$$\mathfrak{m}_{\varphi(P)} / \mathfrak{m}_{\varphi(P)}^2 \longrightarrow \mathfrak{m}_P / \mathfrak{m}_P^2.$$

Therefore the induced morphism on local rings

$$\mathcal{O}_{\mathbb{P}_k^n, \varphi(P)} \longrightarrow \mathcal{O}_{X, P}$$

is surjective.

Hence $\varphi^\#$ is surjective, so φ identifies X with a closed subscheme of $\mathbb{P}_k^n$. Since $\varphi(X)$ is already closed, we conclude that

$$\varphi : X \hookrightarrow \mathbb{P}_k^n$$

is a closed immersion.

Conclusion

Therefore,

$$\varphi \text{ is a closed immersion} \iff V \text{ separates points and tangent vectors.}$$

2 Problem 1 Exactness of Global Sections on an Affine Scheme

Let $X = \operatorname{Spec} A$ be affine, and let

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules. We want to show that

$$0 \longrightarrow \Gamma(X, \mathcal{F}') \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{F}'') \longrightarrow 0$$

is exact.

Solution.

Since $X = \operatorname{Spec} A$ is affine and the sheaves are quasi-coherent, there exist A -modules

$$M', \quad M, \quad M''$$

such that

$$\mathcal{F}' \cong \widetilde{M'}, \quad \mathcal{F} \cong \widetilde{M}, \quad \mathcal{F}'' \cong \widetilde{M''}.$$

For an affine scheme, the functor

$$M \longmapsto \widetilde{M}$$

gives an equivalence between A -modules and quasi-coherent sheaves on $\operatorname{Spec} A$. Under this equivalence, exact sequences of quasi-coherent sheaves correspond exactly to exact sequences of A -modules.

Therefore the given exact sequence of sheaves corresponds to an exact sequence of A -modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

Now use the fundamental property of quasi-coherent sheaves on affine schemes:

$$\Gamma(\operatorname{Spec} A, \widetilde{M}) \cong M.$$

Thus

$$\Gamma(X, \mathcal{F}') \cong M', \quad \Gamma(X, \mathcal{F}) \cong M, \quad \Gamma(X, \mathcal{F}'') \cong M''.$$

Taking global sections gives exactly the module sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

Since this sequence is exact, the sequence

$$0 \longrightarrow \Gamma(X, \mathcal{F}') \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{F}'') \longrightarrow 0$$

is exact.

Hence

$$0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0 \text{ is exact.}$$

3 Problem 2 Cohomology Computed by an Acyclic Resolution

Let $\mathcal{F}$ be a sheaf of abelian groups on a topological space X , and suppose we are given a long exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}^0 \xrightarrow{d^0} \mathcal{F}^1 \xrightarrow{d^1} \mathcal{F}^2 \xrightarrow{d^2} \dots$$

Assume further that

$$H^i(X, \mathcal{F}^j) = 0$$

for all $i > 0$ and all $j \geq 0$. We want to prove that

$$H^i(X, \mathcal{F}) = \frac{\ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1}))}{\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i))}$$

for all $i \geq 0$, with the usual interpretation in degree 0.

Solution.

Define sheaves

$$\mathcal{Z}^0 := \mathcal{F}$$

and, for $i \geq 0$,

$$\mathcal{Z}^{i+1} := \ker(d^i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}).$$

Because the original sequence is exact, we have short exact sequences

$$0 \longrightarrow \mathcal{Z}^i \longrightarrow \mathcal{F}^i \longrightarrow \mathcal{Z}^{i+1} \longrightarrow 0$$

for every $i \geq 0$.

Apply the cohomology functor to this short exact sequence. We obtain the long exact sequence

$$0 \rightarrow H^0(X, \mathcal{Z}^i) \rightarrow H^0(X, \mathcal{F}^i) \rightarrow H^0(X, \mathcal{Z}^{i+1}) \rightarrow H^1(X, \mathcal{Z}^i) \rightarrow H^1(X, \mathcal{F}^i) \rightarrow \dots$$

Since each $\mathcal{F}^i$ is acyclic, we have

$$H^q(X, \mathcal{F}^i) = 0 \quad \text{for all } q > 0.$$

Therefore the long exact sequence gives

$$H^q(X, \mathcal{Z}^{i+1}) \cong H^{q+1}(X, \mathcal{Z}^i)$$

for all $q > 0$.

Repeatedly applying this isomorphism gives

$$H^i(X, \mathcal{F}) = H^i(X, \mathcal{Z}^0) \cong H^{i-1}(X, \mathcal{Z}^1) \cong \dots \cong H^1(X, \mathcal{Z}^{i-1}).$$

Now look at the short exact sequence

$$0 \longrightarrow \mathcal{Z}^{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \mathcal{Z}^i \longrightarrow 0.$$

Its cohomology sequence begins

$$0 \rightarrow H^0(X, \mathcal{Z}^{i-1}) \rightarrow H^0(X, \mathcal{F}^{i-1}) \rightarrow H^0(X, \mathcal{Z}^i) \rightarrow H^1(X, \mathcal{Z}^{i-1}) \rightarrow H^1(X, \mathcal{F}^{i-1}).$$

Since

$$H^1(X, \mathcal{F}^{i-1}) = 0,$$

we get an exact sequence

$$H^0(X, \mathcal{F}^{i-1}) \rightarrow H^0(X, \mathcal{Z}^i) \rightarrow H^1(X, \mathcal{Z}^{i-1}) \rightarrow 0.$$

Hence

$$H^1(X, \mathcal{Z}^{i-1}) \cong \frac{H^0(X, \mathcal{Z}^i)}{\operatorname{im}(H^0(X, \mathcal{F}^{i-1}) \rightarrow H^0(X, \mathcal{Z}^i))}.$$

Since

$$H^0(X, \mathcal{Z}^i) = \Gamma(X, \mathcal{Z}^i),$$

and

$$\mathcal{Z}^i = \ker(d^i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}),$$

we have

$$\Gamma(X, \mathcal{Z}^i) = \ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1})).$$

Also, the image of

$$\Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{Z}^i)$$

is precisely

$$\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i)).$$

Therefore

$$H^i(X, \mathcal{F}) \cong \frac{\ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1}))}{\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i))}.$$

Thus

$$\boxed{H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{F}^\bullet)).}$$

Equivalently,

$$\boxed{H^i(X, \mathcal{F}) = \frac{\ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1}))}{\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i))}.$$

This proves that an acyclic resolution computes sheaf cohomology.

4 Theory: Closed Immersions, Affine Schemes, and Acyclic Resolutions

4.1 Linear Systems and Closed Immersions

Let X be a scheme and let $\mathcal{L}$ be an invertible sheaf on X . A finite-dimensional subspace

$$V \subseteq \Gamma(X, \mathcal{L})$$

is called a linear system. If V is generated by sections

$$s_0, \dots, s_n,$$

and these sections generate $\mathcal{L}$, then they define a morphism

$$\varphi : X \rightarrow \mathbb{P}^n.$$

At a point $P \in X$, the image is given by

$$P \mapsto [s_0(P) : \dots : s_n(P)].$$

The map φ is a closed immersion precisely when the sections contain enough information to recover both:

points of X ,

and

infinitesimal tangent directions of X .

This leads to two conditions.

First, V must separate points. This means that for distinct closed points

$$P \neq Q,$$

there is a section $s \in V$ that vanishes at one point but not at the other. Geometrically, this prevents two different points from being mapped to the same point of projective space.

Second, V must separate tangent vectors. At a closed point P , the cotangent space is

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

The sections of V vanishing at P should generate this space. This means that the morphism does not collapse first-order infinitesimal directions.

Thus, the criterion is:

$$\boxed{\varphi : X \rightarrow \mathbb{P}^n \text{ is a closed immersion} \iff V \text{ separates points and tangent vectors.}}$$

The condition of separating points controls the underlying topological embedding, while separating tangent vectors controls the infinitesimal scheme structure.

4.2 Quasi-Coherent Sheaves on Affine Schemes

Let

$$X = \operatorname{Spec} A.$$

One of the fundamental facts about affine schemes is that quasi-coherent sheaves on X are equivalent to A -modules.

More precisely, if M is an A -module, then one obtains a quasi-coherent sheaf

$$\widetilde{M}$$

on X . Conversely, every quasi-coherent sheaf on X is of this form.

Moreover,

$$\Gamma(X, \widetilde{M}) \cong M.$$

Therefore, quasi-coherent sheaves on affine schemes behave very much like modules. In particular, exact sequences of quasi-coherent sheaves correspond to exact sequences of modules.

So if

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

is an exact sequence of quasi-coherent sheaves on an affine scheme, then it corresponds to an exact sequence of A -modules

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0.$$

Taking global sections gives

$$0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0,$$

which is exact.

Hence on affine schemes, the global section functor is exact on quasi-coherent sheaves:

$X = \operatorname{Spec} A, \mathcal{F} \text{ quasi-coherent} \implies \Gamma(X, -) \text{ is exact on quasi-coherent sheaves.}$

This is one of the main reasons affine schemes are algebraically manageable.

4.3 Acyclic Sheaves and Acyclic Resolutions

For a sheaf $\mathcal{G}$ on X , the groups

$$H^i(X, \mathcal{G})$$

measure the failure of global sections to be exact.

A sheaf $\mathcal{G}$ is called acyclic if

$$H^i(X, \mathcal{G}) = 0 \quad \text{for all } i > 0.$$

Examples of acyclic sheaves often include injective sheaves, flasque sheaves, and quasi-coherent sheaves on affine schemes.

Suppose we have an exact resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}^0 \xrightarrow{d^0} \mathcal{F}^1 \xrightarrow{d^1} \mathcal{F}^2 \rightarrow \dots$$

where every $\mathcal{F}^j$ is acyclic.

Then this is called an acyclic resolution of $\mathcal{F}$.

The key theorem says that cohomology of $\mathcal{F}$ can be computed by applying global sections to the resolution.

That is,

$$H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{F}^\bullet)).$$

Explicitly,

$$H^i(X, \mathcal{F}) = \frac{\ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1}))}{\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i))}.$$

This theorem is extremely important because it allows one to replace the difficult problem of computing sheaf cohomology by the more concrete problem of computing the cohomology of a complex of abelian groups.

4.4 Relationship Between the Three Results

The three problems fit together naturally.

The closed immersion criterion studies when a morphism into projective space captures all geometric information of a scheme. It focuses on points and tangent directions.

The exactness result for quasi-coherent sheaves on affine schemes says that affine schemes have no higher obstruction to taking global sections. Algebraically, everything is controlled by modules.

The acyclic resolution theorem generalizes this idea. If a sheaf is resolved by sheaves whose higher cohomology vanishes, then its cohomology can be computed directly from global sections.

Together, these results express a central theme of algebraic geometry:

Geometry is controlled by functions, local algebra, and cohomology.

More specifically:

Topic	Main object	Main idea
Closed immersion	linear system V	sections separate points and tangent vectors
Affine exactness	quasi-coherent sheaves	global sections behave like modules
Acyclic resolution	sheaf cohomology	cohomology is computed by global sections

5 Theory: Sections, Affine Schemes, and Acyclic Resolutions

The three results are connected by one common principle: global sections can encode geometric information, algebraic information, and cohomological information, provided the sheaves involved have good properties.

Let $\mathcal{L}$ be an invertible sheaf on a scheme X , and let

$$V = \langle s_0, \dots, s_n \rangle \subseteq \Gamma(X, \mathcal{L})$$

be a finite-dimensional space of global sections generating $\mathcal{L}$. These sections define a morphism

$$\varphi : X \rightarrow \mathbb{P}_k^n$$

by

$$P \mapsto [s_0(P) : \dots : s_n(P)].$$

For this morphism to be a closed immersion, the sections must distinguish both ordinary points and infinitesimal directions. Separating points means that for any two distinct closed points $P, Q \in X$, there exists a section $s \in V$ which vanishes at one of the points but not at the other. This prevents φ from identifying different points of X . Separating tangent vectors means that, for each closed point $P \in X$, the sections of V vanishing at P generate

$$\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P.$$

After trivializing $\mathcal{L}$ near P , this is the cotangent space

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

Thus the morphism also preserves first-order infinitesimal information. Hence one has the criterion

$$\boxed{\varphi : X \rightarrow \mathbb{P}_k^n \text{ is a closed immersion} \iff V \text{ separates points and tangent vectors.}}$$

The second result concerns affine schemes. If

$$X = \operatorname{Spec} A,$$

then quasi-coherent sheaves on X are equivalent to A -modules. For every A -module M , there is an associated quasi-coherent sheaf $\widetilde{M}$, and every quasi-coherent sheaf on X is of this form. Moreover,

$$\Gamma(X, \widetilde{M}) \cong M.$$

Therefore global sections recover the original module. Consequently, an exact sequence of quasi-coherent sheaves

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

on an affine scheme corresponds to an exact sequence of A -modules

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0.$$

Taking global sections gives exactly this module sequence:

$$0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0.$$

Thus, on affine schemes, the global section functor is exact on quasi-coherent sheaves:

$$\boxed{X = \operatorname{Spec} A \implies \Gamma(X, -) \text{ is exact on quasi-coherent sheaves.}}$$

The third result places this exactness into the broader framework of sheaf cohomology. For a sheaf $\mathcal{G}$, the groups

$$H^i(X, \mathcal{G})$$

measure the failure of global sections to be exact. A sheaf $\mathcal{G}$ is called acyclic if

$$H^i(X, \mathcal{G}) = 0 \quad \text{for all } i > 0.$$

Suppose we have an exact resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}^0 \xrightarrow{d^0} \mathcal{F}^1 \xrightarrow{d^1} \mathcal{F}^2 \rightarrow \dots$$

where every $\mathcal{F}^j$ is acyclic. Then the cohomology of $\mathcal{F}$ can be computed by applying global sections to the resolution:

$$H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{F}^\bullet)).$$

Explicitly,

$$H^i(X, \mathcal{F}) = \frac{\ker(d^i : \Gamma(X, \mathcal{F}^i) \rightarrow \Gamma(X, \mathcal{F}^{i+1}))}{\operatorname{im}(d^{i-1} : \Gamma(X, \mathcal{F}^{i-1}) \rightarrow \Gamma(X, \mathcal{F}^i))}.$$

Thus the three results show three levels at which global sections are useful. In projective geometry, sections define morphisms and embeddings. In affine geometry, global sections translate quasi-coherent sheaves into modules. In cohomology, acyclic resolutions allow global sections to compute derived invariants. The common theme is

geometry is encoded by sections, local algebra, and cohomology.

6 Examples: Closed Immersions, Affine Exactness, and Acyclic Resolutions

Example 1: A linear system separating points on $\mathbb{P}^1$

Let

$$X = \mathbb{P}_k^1$$

and let

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1).$$

The global sections are generated by

$$x_0, x_1.$$

They define the identity embedding

$$\mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1, \quad P \mapsto [x_0(P) : x_1(P)].$$

The sections x_0, x_1 separate points because two distinct points of $\mathbb{P}^1$ have different homogeneous coordinates. They also separate tangent vectors, since $\mathcal{O}_{\mathbb{P}^1}(1)$ gives the usual projective embedding.

Therefore

$$\mathbb{P}_k^1 \hookrightarrow \mathbb{P}_k^1$$

is a closed immersion.

Example 2: The Veronese embedding

Let

$$X = \mathbb{P}_k^1, \quad \mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2).$$

The global sections are

$$x_0^2, \quad x_0x_1, \quad x_1^2.$$

They define

$$\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^2$$

by

$$[x_0 : x_1] \mapsto [x_0^2 : x_0x_1 : x_1^2].$$

This is the degree two Veronese embedding. Its image is the conic

$$Y^2 = XZ$$

inside $\mathbb{P}_k^2$.

The sections separate points and tangent vectors, so

$$\varphi : \mathbb{P}_k^1 \hookrightarrow \mathbb{P}_k^2$$

is a closed immersion.

Example 3: A map that does not separate points

Consider

$$\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$$

defined by

$$[x_0 : x_1] \mapsto [x_0^2 : x_1^2].$$

If $\text{char}(k) \neq 2$, then

$$[x_0 : x_1] \quad \text{and} \quad [x_0 : -x_1]$$

have the same image, because

$$[x_0^2 : x_1^2] = [x_0^2 : (-x_1)^2].$$

Thus the corresponding sections do not separate points. Therefore this morphism is not a closed immersion.

Example 4: Quasi-coherent sheaves on an affine scheme

Let

$$X = \text{Spec } k[x].$$

Let

$$A = k[x].$$

Take the exact sequence of A -modules

$$0 \rightarrow (x) \rightarrow A \rightarrow A/(x) \rightarrow 0.$$

Passing to associated sheaves gives

$$0 \rightarrow \widetilde{(x)} \rightarrow \widetilde{A} \rightarrow \widetilde{A/(x)} \rightarrow 0.$$

Since X is affine, taking global sections gives back the original exact sequence:

$$0 \rightarrow \Gamma(X, \widetilde{(x)}) \rightarrow \Gamma(X, \widetilde{A}) \rightarrow \Gamma(X, \widetilde{A/(x)}) \rightarrow 0.$$

But

$$\Gamma(X, \widetilde{(x)}) = (x), \quad \Gamma(X, \widetilde{A}) = A, \quad \Gamma(X, \widetilde{A/(x)}) = A/(x).$$

Hence

$$0 \rightarrow (x) \rightarrow k[x] \rightarrow k[x]/(x) \rightarrow 0$$

is exact.

Example 5: The skyscraper sheaf at a point

Let

$$X = \operatorname{Spec} k[x],$$

and let $P = (x)$. The quotient

$$k[x]/(x) \cong k$$

corresponds to the skyscraper sheaf k_P supported at the point P .

Thus

$$\widetilde{k[x]/(x)} \cong k_P.$$

The exact sequence

$$0 \rightarrow (x) \rightarrow k[x] \rightarrow k[x]/(x) \rightarrow 0$$

becomes

$$0 \rightarrow \widetilde{(x)} \rightarrow \mathcal{O}_X \rightarrow k_P \rightarrow 0.$$

Taking global sections remains exact:

$$0 \rightarrow (x) \rightarrow k[x] \rightarrow k \rightarrow 0.$$

This illustrates that on affine schemes, even quotient sheaves behave algebraically like quotient modules.

Example 6: Acyclic resolution by flasque sheaves

Let $\mathcal{F}$ be any sheaf of abelian groups on a topological space X . One can embed $\mathcal{F}$ into a flasque sheaf:

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{C}^0.$$

Then continue by taking the cokernel:

$$\mathcal{Q}^1 = \text{coker}(\mathcal{F} \rightarrow \mathcal{C}^0),$$

and embed again:

$$0 \rightarrow \mathcal{Q}^1 \rightarrow \mathcal{C}^1.$$

Repeating this process gives a flasque resolution

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{C}^0 \rightarrow \mathcal{C}^1 \rightarrow \mathcal{C}^2 \rightarrow \cdots.$$

Flasque sheaves are acyclic, so

$$H^i(X, \mathcal{C}^j) = 0 \quad (i > 0).$$

Therefore

$$H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{C}^\bullet)).$$

This is one of the standard ways to compute sheaf cohomology abstractly.

Example 7: Affine schemes have no higher cohomology for quasi-coherent sheaves

Let

$$X = \text{Spec } A$$

and let $\mathcal{F} = \widetilde{M}$ be quasi-coherent.

Then

$$H^i(X, \mathcal{F}) = 0 \quad \text{for all } i > 0.$$

For example, for

$$X = \text{Spec } k[x, y], \quad \mathcal{F} = \widetilde{k[x, y]/(x, y)},$$

we have

$$H^1(X, \mathcal{F}) = 0, \quad H^2(X, \mathcal{F}) = 0, \quad \dots$$

This is the cohomological reason why affine schemes are algebraically simple.

Example 8: Computing cohomology from an acyclic resolution

Suppose

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}^0 \rightarrow \mathcal{F}^1 \rightarrow \mathcal{F}^2 \rightarrow \dots$$

is an acyclic resolution.

After taking global sections, we obtain a complex

$$0 \rightarrow \Gamma(X, \mathcal{F}^0) \rightarrow \Gamma(X, \mathcal{F}^1) \rightarrow \Gamma(X, \mathcal{F}^2) \rightarrow \dots$$

Then

$$H^0(X, \mathcal{F}) = \ker(\Gamma(X, \mathcal{F}^0) \rightarrow \Gamma(X, \mathcal{F}^1)),$$

and

$$H^1(X, \mathcal{F}) = \frac{\ker(\Gamma(X, \mathcal{F}^1) \rightarrow \Gamma(X, \mathcal{F}^2))}{\operatorname{im}(\Gamma(X, \mathcal{F}^0) \rightarrow \Gamma(X, \mathcal{F}^1))}.$$

Similarly,

$$H^2(X, \mathcal{F}) = \frac{\ker(\Gamma(X, \mathcal{F}^2) \rightarrow \Gamma(X, \mathcal{F}^3))}{\operatorname{im}(\Gamma(X, \mathcal{F}^1) \rightarrow \Gamma(X, \mathcal{F}^2))}.$$

Thus sheaf cohomology is computed exactly like ordinary cohomology of a complex.

Summary Table

Example	Main object	Main lesson
$\mathbb{P}^1 \hookrightarrow \mathbb{P}^1$	$\mathcal{O}(1)$	sections separate points and tangents
$\mathbb{P}^1 \hookrightarrow \mathbb{P}^2$	$\mathcal{O}(2)$	Veronese map is a closed immersion
$[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$	linear system	failure to separate points
$\operatorname{Spec} k[x]$	quasi-coherent sheaves	$\Gamma(X, -)$ is exact
$0 \rightarrow \mathcal{Y} \rightarrow k[x] \rightarrow k \rightarrow 0$	modules and sheaves	exactness is preserved
flasque resolution	acyclic sheaves	computes sheaf cohomology
$\operatorname{Spec} A$	quasi-coherent sheaf	$H^i(X, \mathcal{F}) = 0$ for $i > 0$

7 Diagrams, Charts, and Summary Tables

7.1 Diagram 1: Closed Immersion Criterion

Let $\mathcal{V} = \langle s_0, \dots, s_n \rangle$ be a linear system defining a morphism

$$\varphi_{\mathcal{V}} : X \longrightarrow \mathbb{P}^n.$$

7.2 Diagram 1: Closed Immersion Criterion

Let $\mathcal{V} = \langle s_0, \dots, s_n \rangle$ be a linear system defining a morphism

$$\varphi_{\mathcal{V}} : X \longrightarrow \mathbb{P}^n.$$

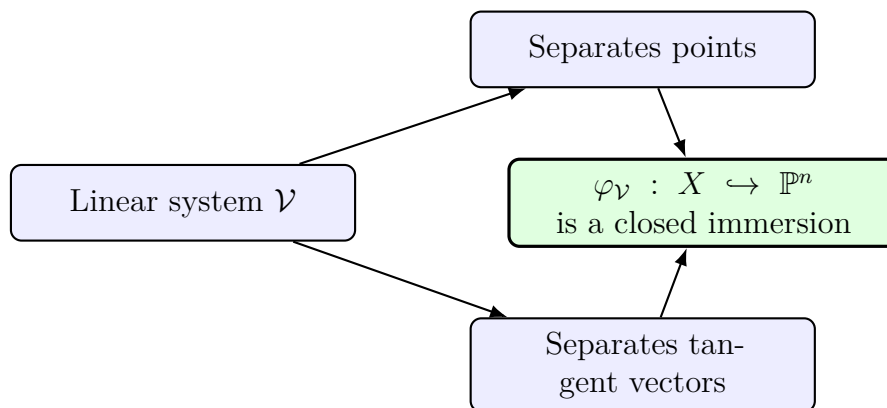
$$X \xrightarrow{\varphi_{\mathcal{V}}} \mathbb{P}^n$$

$$P \longmapsto [s_0(P) : \cdots : s_n(P)]$$

$$\boxed{\varphi_{\mathcal{V}} \text{ is a closed immersion}}$$

if and only if the linear system satisfies both conditions:

$$\begin{cases} \text{its sections separate points,} \\ \text{its sections separate tangent vectors.} \end{cases}$$



7.3 Diagram 2: Separating Points vs Tangent Directions

point separation

$$\begin{array}{ccc} & \xrightarrow{\quad} & \\ \bullet & & \bullet \\ P & & Q \end{array}$$

$$\begin{array}{ccc} & v_1 & \\ & \nearrow & \\ P & & \\ & \searrow & \\ & v_2 & \end{array}$$

The first picture represents separating points. The second represents separating tangent vectors.

“`latex`

7.4 Diagram 3: Affine Scheme and Global Sections

Let

$$X = \operatorname{Spec}(A).$$

$$A\text{-}\mathbf{Mod} \begin{array}{c} \xrightarrow{M \mapsto \widetilde{M}} \\ \xleftarrow{\Gamma(X, -)} \end{array} \mathbf{QCoh}(X)$$

The two categories are equivalent. Moreover,

$$\Gamma(X, \widetilde{M}) \cong M.$$

7.5 Diagram 4: Exactness on Affine Schemes

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

““

7.6 Diagram 4: Exactness on Affine Schemes

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

corresponds to

$$0 \longrightarrow \widetilde{M}' \longrightarrow \widetilde{M} \longrightarrow \widetilde{M}'' \longrightarrow 0$$

and after taking global sections

$$0 \longrightarrow \Gamma(X, \widetilde{M}') \longrightarrow \Gamma(X, \widetilde{M}) \longrightarrow \Gamma(X, \widetilde{M}'') \longrightarrow 0.$$

7.7 Diagram 5: Acyclic Resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \mathcal{F}^2 \longrightarrow \dots$$

with

$$H^i(X, \mathcal{F}^j) = 0 \quad (i > 0).$$

Applying global sections gives

$$0 \longrightarrow \Gamma(X, \mathcal{F}^0) \longrightarrow \Gamma(X, \mathcal{F}^1) \longrightarrow \Gamma(X, \mathcal{F}^2) \longrightarrow \dots$$

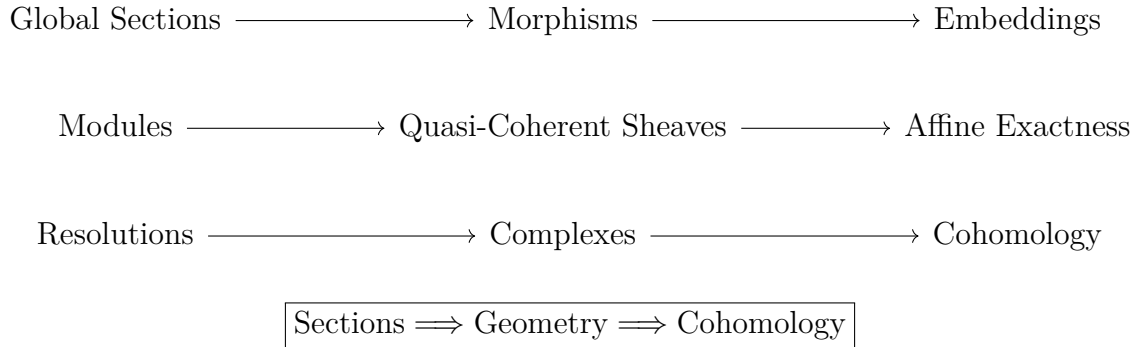
whose cohomology computes

$$H^i(X, \mathcal{F}).$$

7.8 Summary Table

Topic	Input	Condition	Output
Closed immersion	$V \subset \Gamma(X, \mathcal{L})$	Separate points & tangents	$X \hookrightarrow \mathbb{P}^n$
Affine exactness	$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$	$X = \operatorname{Spec} A$	$\Gamma(X, -)$ exact
Acyclic resolution	$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}^\bullet$	$H^{>0}(X, \mathcal{F}^j) = 0$	$H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{F}^\bullet))$

7.9 Big Picture Chart



8 Immersions and Closed Immersions Derived from Linear Systems

Throughout this section, let k be an algebraically closed field. When we discuss the map

$$[s : t] \longmapsto [s^2 : t^2],$$

we additionally assume that $\operatorname{char}(k) \neq 2$.

The purpose of this section is to explain, through explicit computations, how morphisms defined by functions or by sections of line bundles become immersions or closed immersions. The central examples are morphisms from affine and projective spaces, especially those obtained from linear systems on $\mathbb{P}^1$.

8.1 Basic definitions and the main criterion

Definition 8.1.1 (Closed immersion). A morphism of schemes

$$i : Z \longrightarrow X$$

is a *closed immersion* if:

- (i) the induced map of topological spaces identifies Z homeomorphically with a closed subset of X ;
- (ii) the morphism of sheaves of rings

$$\mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z$$

is surjective.

Equivalently, a closed subscheme $Z \subseteq X$ is determined by a quasi-coherent ideal sheaf

$$\mathcal{I}_{Z/X} \subseteq \mathcal{O}_X,$$

and one has the exact sequence

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Definition 8.1.2 (Immersion). A morphism

$$f : Y \longrightarrow X$$

is an *immersion* if it factors as

$$Y \xrightarrow{j} U \xrightarrow{i} X,$$

where j is a closed immersion and $i : U \hookrightarrow X$ is an open immersion.

Thus an immersion identifies Y with a locally closed subscheme of X , while a closed immersion identifies Y with a closed subscheme of X .

Affine algebra criterion

Let

$$\varphi : B \longrightarrow A$$

be a homomorphism of rings. It induces a morphism

$$f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B).$$

The morphism f is a closed immersion if and only if φ is surjective. In that case,

$$A \cong B/I, \quad I = \ker(\varphi),$$

and the image of f is the closed subscheme

$$V(I) \subseteq \operatorname{Spec}(B).$$

$\operatorname{Spec}(B/I) \hookrightarrow \operatorname{Spec}(B)$ is the affine model of a closed immersion.

Criterion for a linear system

Let X be a projective variety and let $\mathcal{L}$ be a line bundle on X . A finite-dimensional subspace

$$V \subseteq H^0(X, \mathcal{L})$$

defines a morphism

$$\varphi_V : X \longrightarrow \mathbb{P}(V^\vee)$$

provided that V is base-point-free.

If

$$V = \langle s_0, \dots, s_n \rangle,$$

then

$$\varphi_V(P) = [s_0(P) : \cdots : s_n(P)].$$

For a projective variety, the morphism φ_V is a closed immersion when the linear system:

$$\begin{cases} \text{separates points,} \\ \text{separates tangent vectors.} \end{cases}$$

More concretely:

- V separates points if for any distinct $P, Q \in X$, there is a section in V distinguishing P from Q ;
- V separates tangent vectors if it detects every nonzero first-order infinitesimal direction at each point.

The examples below show each condition explicitly.

8.2 Step-by-step affine examples

Example 1: A closed point inside the affine line

Consider the origin

$$\{0\} \subseteq \mathbb{A}_k^1 = \text{Spec}(k[x]).$$

Step 1: Write the defining ideal.

The origin is defined by the ideal

$$(x) \subseteq k[x].$$

Step 2: Form the quotient coordinate ring.

We have

$$k[x]/(x) \cong k.$$

Therefore

$$\text{Spec}(k[x]/(x)) \cong \text{Spec}(k).$$

Step 3: Construct the morphism.

The quotient map

$$k[x] \longrightarrow k[x]/(x) \cong k$$

induces

$$\text{Spec}(k) \hookrightarrow \text{Spec}(k[x]) = \mathbb{A}_k^1.$$

Step 4: Conclude closed immersion.

Since the ring homomorphism is surjective, the induced morphism is a closed immersion.

$$\boxed{\text{Spec}(k) = V(x) \hookrightarrow \mathbb{A}_k^1}$$

This is the simplest closed immersion: a single reduced point embedded as a closed subscheme.

Example 2: The parabola as a closed immersion into $\mathbb{A}^2$

Define

$$f : \mathbb{A}_k^1 \longrightarrow \mathbb{A}_k^2$$

by

$$t \longmapsto (t, t^2).$$

We show step by step that f is a closed immersion.

Step 1: Pass to coordinate rings.

Since

$$\mathbb{A}_k^1 = \operatorname{Spec}(k[t]), \quad \mathbb{A}_k^2 = \operatorname{Spec}(k[x, y]),$$

the morphism f corresponds to

$$f^\# : k[x, y] \longrightarrow k[t]$$

defined by

$$x \longmapsto t, \quad y \longmapsto t^2.$$

Step 2: Check surjectivity.

The element $t \in k[t]$ is the image of x . Therefore every polynomial in t lies in the image of $f^\#$. Hence

$$f^\# : k[x, y] \twoheadrightarrow k[t]$$

is surjective.

Step 3: Compute the kernel.

The polynomial

$$y - x^2$$

maps to

$$t^2 - t^2 = 0.$$

Thus

$$(y - x^2) \subseteq \ker(f^\#).$$

Conversely,

$$k[x, y]/(y - x^2) \cong k[x] \cong k[t].$$

Hence

$$\ker(f^\#) = (y - x^2).$$

Step 4: Identify the image.

The image is the closed subscheme

$$V(y - x^2) \subseteq \mathbb{A}_k^2.$$

Thus

$$\mathbb{A}_k^1 \cong V(y - x^2) \hookrightarrow \mathbb{A}_k^2.$$

Conclusion.

$t \mapsto (t, t^2)$ is a closed immersion whose image is the parabola $y = x^2$.

This example illustrates the affine principle: an explicit parametrization is a closed immersion when its coordinate-ring map is surjective.

Example 3: An open immersion that is not a closed immersion

Consider

$$D(x) = \mathbb{A}_k^1 \setminus \{0\}.$$

This open subset is isomorphic to

$$\operatorname{Spec}(k[x, x^{-1}]).$$

The inclusion

$$j : D(x) \hookrightarrow \mathbb{A}_k^1$$

is an open immersion.

Step 1: Identify its image.

The image is

$$\mathbb{A}_k^1 \setminus \{0\},$$

which is open.

Step 2: Ask whether the image is closed.

The set $D(x)$ is not closed in $\mathbb{A}_k^1$, since its closure is all of $\mathbb{A}_k^1$.

Step 3: Conclude.

Therefore

$$D(x) \hookrightarrow \mathbb{A}_k^1$$

is an immersion, because every open immersion is an immersion, but it is not a closed immersion.

$\mathbb{G}_m = \operatorname{Spec}(k[x, x^{-1}]) \hookrightarrow \mathbb{A}_k^1$ is open but not closed.

Example 4: A locally closed immersion

Let

$$X = \mathbb{A}_k^2 = \operatorname{Spec}(k[x, y]).$$

Inside X , consider the subset

$$Y = V(y) \cap D(x).$$

Geometrically, Y is the x -axis with the origin removed.

Step 1: First pass to an open subset.

The distinguished open subset

$$D(x) \subseteq \mathbb{A}_k^2$$

has coordinate ring

$$k[x, y, x^{-1}].$$

Step 2: Impose the closed equation $y = 0$ inside this open subset.

Inside $D(x)$, the subscheme Y is defined by the ideal

$$(y) \subseteq k[x, y, x^{-1}].$$

Thus

$$Y \cong \operatorname{Spec}(k[x, y, x^{-1}]/(y)) \cong \operatorname{Spec}(k[x, x^{-1}]).$$

Step 3: Factor the morphism.

The inclusion factors as

$$Y \hookrightarrow D(x) \hookrightarrow \mathbb{A}_k^2,$$

where the first map is a closed immersion and the second map is an open immersion.

Conclusion.

$V(y) \cap D(x) \hookrightarrow \mathbb{A}_k^2 \text{ is a locally closed immersion.}$

This is the model example of an immersion which is neither merely an open inclusion nor a closed inclusion into the ambient space.

8.3 Linear systems on $\mathbb{P}^1$

Let

$$\mathbb{P}_k^1 = \operatorname{Proj}(k[s, t]).$$

For each $d \geq 0$,

$$H^0(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}^1}(d))$$

is the vector space of homogeneous polynomials of degree d in s, t .

A choice of sections

$$f_0, \dots, f_n \in H^0(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}^1}(d))$$

with no common zero defines

$$[s : t] \longmapsto [f_0(s, t) : \dots : f_n(s, t)].$$

The next examples explain when this morphism is a closed immersion.

Example 5: The complete linear system $\mathcal{O}_{\mathbb{P}^1}(1)$

Consider

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1).$$

Its global sections are

$$H^0(\mathbb{P}^1, \mathcal{O}(1)) = \langle s, t \rangle.$$

Step 1: Construct the morphism.

The complete linear system defines

$$\varphi_{|\mathcal{O}(1)|} : \mathbb{P}^1 \longrightarrow \mathbb{P}^1$$

by

$$[s : t] \longmapsto [s : t].$$

Step 2: Check base-point-freeness.

The sections s and t do not vanish simultaneously at any point of $\mathbb{P}^1$. Therefore the map is defined everywhere.

Step 3: Check separation of points.

If

$$[s_0 : t_0] \neq [s_1 : t_1],$$

then their coordinate pairs determine distinct points of $\mathbb{P}^1$. Hence the sections s, t separate points.

Step 4: Check tangent directions.

On the affine chart $s \neq 0$, write

$$u = \frac{t}{s}.$$

The morphism becomes

$$u \longmapsto u.$$

Its differential is

$$du \longmapsto du,$$

which is nonzero. The same argument holds on the chart $t \neq 0$.

Thus tangent vectors are separated.

Conclusion.

The map is the identity morphism, hence a closed immersion:

$$\boxed{\mathbb{P}^1 \xrightarrow{|\mathcal{O}(1)|} \mathbb{P}^1 \text{ is a closed immersion.}}$$

Example 6: The quadratic Veronese embedding from $\mathcal{O}_{\mathbb{P}^1}(2)$

Now take

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2).$$

Its global sections are

$$H^0(\mathbb{P}^1, \mathcal{O}(2)) = \langle s^2, st, t^2 \rangle.$$

The associated morphism is

$$\nu_2 : \mathbb{P}^1 \longrightarrow \mathbb{P}^2$$

given by

$$[s : t] \longmapsto [s^2 : st : t^2].$$

We verify step by step that this is a closed immersion.

Step 1: Check that the map is defined everywhere.

The three sections

$$s^2, \quad st, \quad t^2$$

cannot all vanish at one point of $\mathbb{P}^1$, because that would imply

$$s = t = 0,$$

which does not define a point of projective space.

Therefore ν_2 is a morphism.

Step 2: Compute an equation satisfied by the image.

Write homogeneous coordinates on $\mathbb{P}^2$ as

$$[X : Y : Z].$$

For a point in the image,

$$X = s^2, \quad Y = st, \quad Z = t^2.$$

Hence

$$XZ - Y^2 = s^2t^2 - (st)^2 = 0.$$

Therefore

$$\text{im}(\nu_2) \subseteq V(XZ - Y^2) \subseteq \mathbb{P}^2.$$

Step 3: Recover the parameter from the image on an affine chart.

On the chart $X \neq 0$, we have $s \neq 0$. Set

$$u = \frac{t}{s}.$$

Then

$$[s : t] = [1 : u],$$

and

$$\nu_2([1 : u]) = [1 : u : u^2].$$

Thus on this chart, the map is

$$u \longmapsto (u, u^2),$$

which is the affine parabola embedding.

Similarly, on the chart $Z \neq 0$, set

$$v = \frac{s}{t}.$$

Then

$$\nu_2([v : 1]) = [v^2 : v : 1].$$

Hence the map is locally a closed immersion on a cover of $\mathbb{P}^1$.

Step 4: Check point separation.

Suppose

$$[s : t] \neq [s' : t'].$$

On either affine chart, the parameter $u = t/s$ or $v = s/t$ is recovered from the image coordinate Y/X or Y/Z . Therefore distinct points have distinct images.

Step 5: Check tangent separation.

On the affine chart $s \neq 0$, the map is

$$u \longmapsto (u, u^2).$$

Its differential is

$$du \longmapsto (du, 2u du).$$

The first component is du , so a nonzero tangent vector never maps to zero. Thus tangent vectors are separated.

Step 6: Identify the image.

The image is precisely the smooth conic

$$V(XZ - Y^2) \subseteq \mathbb{P}^2.$$

Conclusion.

$$\boxed{\nu_2 : \mathbb{P}^1 \hookrightarrow \mathbb{P}^2, \quad [s : t] \longmapsto [s^2 : st : t^2],}$$

is a closed immersion, whose image is the projective conic

$$XZ - Y^2 = 0.$$

This is the quadratic Veronese embedding.

Example 7: A base-point-free linear system that is not an immersion

Consider the two-dimensional subspace

$$V = \langle s^2, t^2 \rangle \subseteq H^0(\mathbb{P}^1, \mathcal{O}(2)).$$

It defines a morphism

$$\psi : \mathbb{P}^1 \longrightarrow \mathbb{P}^1$$

given by

$$[s : t] \longmapsto [s^2 : t^2].$$

Step 1: Check base-point-freeness.

The sections s^2 and t^2 do not vanish simultaneously anywhere on $\mathbb{P}^1$. Hence ψ is a morphism.

Step 2: Compare two distinct points.

Since $\text{char}(k) \neq 2$, the points

$$[1 : 1] \quad \text{and} \quad [1 : -1]$$

are distinct in $\mathbb{P}^1$.

But

$$\psi([1 : 1]) = [1 : 1]$$

and

$$\psi([1 : -1]) = [1 : (-1)^2] = [1 : 1].$$

Thus ψ does not separate points.

Step 3: Conclude failure of immersion.

A closed immersion must be injective on points. Since ψ identifies two distinct points, it is not a closed immersion and not even an immersion.

$[s : t] \longmapsto [s^2 : t^2] \text{ is base-point-free but is not an immersion.}$

Lesson.

Base-point-freeness guarantees only that a linear system defines a morphism. It does not guarantee that the morphism is an embedding. To obtain a closed immersion, the linear system must also separate points and tangent vectors.

Example 8: A linear system with a base point gives only a rational map

Let

$$X = \mathbb{P}_k^2$$

with homogeneous coordinates

$$[x : y : z].$$

Consider the subspace

$$V = \langle x, y \rangle \subseteq H^0(\mathbb{P}^2, \mathcal{O}(1)).$$

Formally, these sections suggest a map

$$[x : y : z] \dashrightarrow [x : y] \in \mathbb{P}^1.$$

Step 1: Locate the common zero locus.

The sections x and y vanish simultaneously at

$$[0 : 0 : 1].$$

Thus V has a base point.

Step 2: Observe failure of definition.

At the point $[0 : 0 : 1]$, the expression

$$[x : y] = [0 : 0]$$

does not define a point of $\mathbb{P}^1$.

Step 3: Conclude.

The linear system does not define a morphism on all of $\mathbb{P}^2$. It defines only a rational map

$$\mathbb{P}^2 \dashrightarrow \mathbb{P}^1,$$

defined away from the base point.

A linear system with base points may define a rational map, but not a global morphism.

8.4 The general Veronese embedding

For every integer $d \geq 1$, the complete linear system

$$|\mathcal{O}_{\mathbb{P}^1}(d)|$$

has basis

$$s^d, s^{d-1}t, s^{d-2}t^2, \dots, st^{d-1}, t^d.$$

It defines the morphism

$$\nu_d : \mathbb{P}^1 \longrightarrow \mathbb{P}^d$$

given by

$$[s : t] \longmapsto [s^d : s^{d-1}t : s^{d-2}t^2 : \dots : st^{d-1} : t^d].$$

Step 1: Base-point-freeness.

The first and last sections are

$$s^d \quad \text{and} \quad t^d.$$

They cannot vanish simultaneously on $\mathbb{P}^1$. Hence ν_d is defined everywhere.

Step 2: Affine description.

On the chart $s \neq 0$, put

$$u = \frac{t}{s}.$$

Then

$$\nu_d([1 : u]) = [1 : u : u^2 : \dots : u^d].$$

Thus the map locally records the parameter u in its second coordinate and therefore separates both points and tangent directions.

Step 3: Geometric image.

The image is the rational normal curve of degree d in $\mathbb{P}^d$.

Conclusion.

$$\nu_d : \mathbb{P}^1 \hookrightarrow \mathbb{P}^d \text{ is a closed immersion for every } d \geq 1.$$

For $d = 1$, this is the identity embedding of $\mathbb{P}^1$. For $d = 2$, this is the conic embedding. For $d = 3$, this is the twisted cubic embedding

$$[s : t] \longmapsto [s^3 : s^2t : st^2 : t^3] \subseteq \mathbb{P}^3.$$

Linear system	Associated map	Step-by-step issue	Conclusion
$\langle s, t \rangle \subseteq H^0(\mathbb{P}^1, \mathcal{O}(1))$	$[s : t] \mapsto [s : t]$	No base points; separates points and tangent vectors	Closed immersion
$\langle s^2, st, t^2 \rangle \subseteq H^0(\mathbb{P}^1, \mathcal{O}(2))$	$[s : t] \mapsto [s^2 : st : t^2]$	Image satisfies $XZ - Y^2 = 0$; affine charts recover the parameter	Closed immersion onto a conic
$\langle s^2, t^2 \rangle \subseteq H^0(\mathbb{P}^1, \mathcal{O}(2))$	$[s : t] \mapsto [s^2 : t^2]$	No base points, but $[1 : 1]$ and $[1 : -1]$ have the same image	Not an immersion
$\langle x, y \rangle \subseteq H^0(\mathbb{P}^2, \mathcal{O}(1))$	$[x : y : z] \dashrightarrow [x : y]$	The sections vanish simultaneously at $[0 : 0 : 1]$	Only a rational map

Table 1: How base-point-freeness and separation properties determine the geometry of maps induced by linear systems.

8.5 Comparison table: from linear systems to embeddings

8.6 Summary of the step-by-step method

When a morphism is presented by affine equations or by a linear system, the following workflow is useful.

Affine morphisms

Given

$$f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B),$$

proceed as follows:

Step 1. Write the corresponding ring map

$$B \longrightarrow A.$$

Step 2. Check whether this ring map is surjective.

Step 3. If it is surjective, compute its kernel I .

Step 4. Identify the image as

$$V(I) \subseteq \operatorname{Spec}(B).$$

Step 5. Conclude that

$$\operatorname{Spec}(A) \cong V(I) \hookrightarrow \operatorname{Spec}(B)$$

is a closed immersion.

Morphisms defined by linear systems

Given sections

$$s_0, \dots, s_n \in H^0(X, \mathcal{L}),$$

proceed as follows:

Step 1. Check whether the sections have a common zero.

If they do, the expression defines at most a rational map.

Step 2. If there is no common zero, form the morphism

$$\varphi : X \longrightarrow \mathbb{P}^n, \quad P \longmapsto [s_0(P) : \dots : s_n(P)].$$

Step 3. Check whether distinct points of X have distinct images.

Step 4. Check whether nonzero tangent vectors remain visible under the map.

Step 5. If both separation properties hold, conclude that φ is a closed immersion.

The guiding principle is:

surjective coordinate-ring map $\implies$ affine closed immersion,
 base-point-free linear system $\implies$ projective morphism,
 point and tangent separation $\implies$ projective closed immersion.

9 Exact Sequences, Quasi-Coherent Sheaves, and Global Sections on Affine Schemes

Throughout this section, let

$$X = \operatorname{Spec}(A)$$

be an affine scheme. The fundamental principle is that quasi-coherent sheaves on X are governed by A -modules.

If M is an A -module, then its associated quasi-coherent sheaf is denoted

$$\widetilde{M}.$$

Conversely, if $\mathcal{F}$ is a quasi-coherent sheaf on X , then it is recovered from its module of global sections:

$$\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}.$$

Thus affine geometry transforms problems about sheaves into problems about modules.

On an affine scheme, quasi-coherent sheaves are the geometric form of modules.

9.1 The affine correspondence

Let A be a ring and let

$$X = \operatorname{Spec}(A).$$

For every A -module M , one defines a sheaf

$$\widetilde{M}$$

on X by requiring that for every distinguished open subset

$$D(f) \subseteq X,$$

one has

$$\Gamma(D(f), \widetilde{M}) \cong M_f.$$

Here M_f is the localization of M obtained by inverting powers of f .

In particular, for $f = 1$, since $D(1) = X$, we obtain

$$\Gamma(X, \widetilde{M}) \cong M.$$

Theorem 9.1 (Affine correspondence). *Let*

$$X = \operatorname{Spec}(A).$$

Then the functors

$$A\text{-}\mathbf{Mod} \longrightarrow \operatorname{QCoh}(X), \quad M \longmapsto \widetilde{M},$$

and

$$\operatorname{QCoh}(X) \longrightarrow A\text{-}\mathbf{Mod}, \quad \mathcal{F} \longmapsto \Gamma(X, \mathcal{F}),$$

are quasi-inverse equivalences of categories.

This equivalence can be summarized by the diagram

$$A\text{-}\mathbf{Mod} \begin{array}{c} \xrightarrow{M \mapsto \widetilde{M}} \\ \xleftarrow{\Gamma(X, -)} \end{array} \operatorname{QCoh}(X)$$

with

$$\Gamma(X, \widetilde{M}) \cong M$$

for every A -module M , and

$$\widetilde{\Gamma(X, \mathcal{F})} \cong \mathcal{F}$$

for every quasi-coherent sheaf $\mathcal{F}$ on X .

9.2 Quasi-coherent sheaves on distinguished open subsets

The affine correspondence is not only a statement about global sections. It also controls sections over standard affine open subsets.

Let

$$f \in A.$$

The distinguished open subset

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) : f \notin \mathfrak{p}\}$$

is affine:

$$D(f) \cong \operatorname{Spec}(A_f).$$

For an A -module M , the associated sheaf satisfies

$$\Gamma(D(f), \widetilde{M}) \cong M_f.$$

This means that restricting a quasi-coherent sheaf to $D(f)$ corresponds algebraically to localizing its module at f .

$$\boxed{\widetilde{M}|_{D(f)} \longleftrightarrow M_f.}$$

Example: Restricting the structure sheaf of the affine line

Let

$$X = \mathbb{A}_k^1 = \operatorname{Spec}(k[x]).$$

The structure sheaf is

$$\mathcal{O}_X = \widetilde{k[x]}.$$

Consider the distinguished open subset

$$D(x) = \mathbb{A}_k^1 \setminus \{0\}.$$

Then

$$\Gamma(D(x), \mathcal{O}_X) \cong k[x]_x = k[x, x^{-1}].$$

Thus regular functions on the affine line with the origin removed may contain negative powers of x , because x is invertible on $D(x)$.

$$\boxed{\Gamma(\mathbb{A}_k^1 \setminus \{0\}, \mathcal{O}) = k[x, x^{-1}].}$$

9.3 Exact sequences of modules and exact sequences of sheaves

Suppose that

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

is a short exact sequence of A -modules.

Since localization is exact, for every $f \in A$ the sequence

$$0 \longrightarrow M'_f \longrightarrow M_f \longrightarrow M''_f \longrightarrow 0$$

is exact.

But

$$M'_f \cong \Gamma(D(f), \widetilde{M'}), \quad M_f \cong \Gamma(D(f), \widetilde{M}), \quad M''_f \cong \Gamma(D(f), \widetilde{M''}).$$

Therefore the associated sequence of quasi-coherent sheaves

$$0 \longrightarrow \widetilde{M'} \longrightarrow \widetilde{M} \longrightarrow \widetilde{M''} \longrightarrow 0$$

is exact.

Theorem 9.2 (Exactness of the tilde construction). *Let*

$$X = \operatorname{Spec}(A).$$

If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is exact in $A\text{-Mod}$, then

$$0 \longrightarrow \widetilde{M'} \longrightarrow \widetilde{M} \longrightarrow \widetilde{M''} \longrightarrow 0$$

is exact in $\operatorname{QCoh}(X)$.

In fact, because the affine correspondence is an equivalence of categories, the converse also holds: exact sequences of quasi-coherent sheaves on an affine scheme correspond exactly to exact sequences of modules.

Exact sequences of quasi-coherent sheaves on $\operatorname{Spec}(A)$ are exact sequences of A -modules.

9.4 Global sections are exact on affine quasi-coherent sheaves

For arbitrary sheaves on arbitrary spaces, the global-section functor

$$\Gamma(X, -)$$

is only left exact.

However, the situation improves dramatically for quasi-coherent sheaves on an affine scheme.

Theorem 9.3 (Exactness of affine global sections). *Let*

$$X = \operatorname{Spec}(A)$$

be affine. If

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of quasi-coherent sheaves on X , then

$$0 \longrightarrow \Gamma(X, \mathcal{F}') \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{F}'') \longrightarrow 0$$

is an exact sequence of A -modules.

Reason. Write

$$\mathcal{F}' \cong \widetilde{M'}, \quad \mathcal{F} \cong \widetilde{M}, \quad \mathcal{F}'' \cong \widetilde{M''}.$$

The sheaf sequence corresponds to an exact module sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

Since

$$\Gamma(X, \widetilde{M'}) \cong M', \quad \Gamma(X, \widetilde{M}) \cong M, \quad \Gamma(X, \widetilde{M''}) \cong M'',$$

taking global sections simply recovers the original exact module sequence.

Thus:

$\Gamma(X, -)$ is exact on quasi-coherent sheaves when X is affine.

This is equivalent to the vanishing theorem

$$H^i(X, \mathcal{F}) = 0 \quad \text{for every quasi-coherent } \mathcal{F} \text{ and every } i > 0.$$

9.5 Step-by-step examples

Example 1: The origin in $\mathbb{A}_k^1$

Let

$$A = k[x], \quad X = \operatorname{Spec}(A) = \mathbb{A}_k^1.$$

The origin is the closed subscheme

$$Z = V(x) \subseteq X.$$

Its coordinate ring is

$$A/(x) \cong k.$$

Step 1: Start with the module sequence. Multiplication by x gives the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow A/(x) \longrightarrow 0.$$

The first map is injective because

$$A = k[x]$$

is an integral domain and $x \neq 0$.

Step 2: Sheafify the sequence. Applying the tilde construction gives

$$0 \longrightarrow \tilde{A} \xrightarrow{\cdot x} \tilde{A} \longrightarrow \widetilde{A/(x)} \longrightarrow 0.$$

Since

$$\tilde{A} = \mathcal{O}_{\mathbb{A}_k^1}$$

and

$$\widetilde{A/(x)} \cong i_* \mathcal{O}_Z,$$

where $i : Z \hookrightarrow X$, this becomes

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}_k^1} \xrightarrow{\cdot x} \mathcal{O}_{\mathbb{A}_k^1} \longrightarrow i_* \mathcal{O}_Z \longrightarrow 0.$$

Step 3: Interpret the quotient geometrically. The quotient sheaf is supported only at the origin. Algebraically,

$$A/(x) \cong k,$$

so Z is a reduced point.

Thus the sheaf sequence is the ideal-sheaf sequence

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z \longrightarrow 0,$$

with

$$\mathcal{I}_{Z/X} \cong \mathcal{O}_X$$

under the identification given by multiplication by x .

Step 4: Take global sections. Since X is affine and all sheaves are quasi-coherent, global sections preserve exactness:

$$0 \longrightarrow k[x] \xrightarrow{\cdot x} k[x] \longrightarrow k \longrightarrow 0.$$

The final map is evaluation at the origin:

$$f(x) \longmapsto f(0).$$

Conclusion.

$$0 \rightarrow \mathcal{O}_{\mathbb{A}^1} \xrightarrow{\cdot x} \mathcal{O}_{\mathbb{A}^1} \rightarrow \mathcal{O}_{\{0\}} \rightarrow 0$$

is the geometric version of

$$0 \rightarrow k[x] \xrightarrow{\cdot x} k[x] \rightarrow k \rightarrow 0.$$

Example 2: A double point in $\mathbb{A}_k^1$

Let

$$A = k[x], \quad X = \mathbb{A}_k^1.$$

Consider the closed subscheme

$$Z = V(x^2).$$

Although its underlying topological support is still the origin, its coordinate ring is

$$A/(x^2) = k[x]/(x^2),$$

which contains the nonzero nilpotent class of x .

Step 1: Write the exact module sequence. Multiplication by x^2 gives

$$0 \longrightarrow A \xrightarrow{\cdot x^2} A \longrightarrow A/(x^2) \longrightarrow 0.$$

Step 2: Pass to sheaves. Sheafification gives

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\cdot x^2} \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Step 3: Analyze the quotient. As a k -vector space,

$$k[x]/(x^2) = k \oplus kx,$$

with

$$x^2 = 0.$$

Thus the quotient remembers not only the point $x = 0$, but also a first-order infinitesimal direction.

Step 4: Take global sections. Because X is affine,

$$0 \longrightarrow k[x] \xrightarrow{\cdot x^2} k[x] \longrightarrow k[x]/(x^2) \longrightarrow 0$$

is recovered exactly from the sheaf sequence.

Conclusion.

$V(x)$ and $V(x^2)$ have the same support, but different structure sheaves.

The exact sequence detects the difference:

$$\mathcal{O}_{V(x)} \cong k, \quad \mathcal{O}_{V(x^2)} \cong k[x]/(x^2).$$

Example 3: The parabola as a quotient sheaf

Let

$$B = k[x, y], \quad Y = \operatorname{Spec}(B) = \mathbb{A}_k^2.$$

Consider the parabola

$$Z = V(y - x^2) \subseteq Y.$$

Its coordinate ring is

$$B/(y - x^2).$$

Since

$$B/(y - x^2) \cong k[x],$$

the parabola is isomorphic to $\mathbb{A}_k^1$.

Step 1: Write the module sequence. There is a short exact sequence of B -modules

$$0 \longrightarrow (y - x^2) \longrightarrow B \longrightarrow B/(y - x^2) \longrightarrow 0.$$

Because the principal ideal $(y - x^2)$ is isomorphic to B by multiplication with $y - x^2$, one may also write

$$0 \longrightarrow B \xrightarrow{\cdot(y-x^2)} B \longrightarrow B/(y - x^2) \longrightarrow 0.$$

Step 2: Pass to sheaves. On $Y = \operatorname{Spec}(B)$, the associated exact sequence is

$$0 \longrightarrow \mathcal{O}_Y \xrightarrow{\cdot(y-x^2)} \mathcal{O}_Y \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0,$$

where

$$i : Z \hookrightarrow Y$$

is the closed immersion.

Step 3: Interpret the quotient. The quotient sheaf

$$\mathcal{O}_Y/(y - x^2)\mathcal{O}_Y$$

is precisely the structure sheaf of the parabola inside the affine plane.

Step 4: Take global sections. Since Y is affine,

$$0 \longrightarrow k[x, y] \xrightarrow{\cdot(y-x^2)} k[x, y] \longrightarrow k[x, y]/(y - x^2) \longrightarrow 0$$

is exact.

Using

$$k[x, y]/(y - x^2) \cong k[x],$$

we recover regular functions on the parabola as polynomials in one coordinate.

Conclusion.

$$\mathcal{O}_{V(y-x^2)} \cong \mathcal{O}_{\mathbb{A}^2} / (y - x^2) \mathcal{O}_{\mathbb{A}^2},$$

and the exact sequence of sheaves is exactly the geometric form of the quotient-ring construction.

Example 4: The origin in $\mathbb{A}_k^2$ and the Koszul resolution

Let

$$A = k[x, y], \quad X = \operatorname{Spec}(A) = \mathbb{A}_k^2.$$

The origin is the closed subscheme

$$Z = V(x, y).$$

Its coordinate ring is

$$A/(x, y) \cong k.$$

Because the ideal (x, y) has two generators, the structure sheaf of the origin is naturally described by a two-step resolution.

Step 1: Write the Koszul complex of modules. Consider

$$0 \longrightarrow A \xrightarrow{\alpha} A^{\oplus 2} \xrightarrow{\beta} A \longrightarrow A/(x, y) \longrightarrow 0,$$

where

$$\alpha(c) = (-yc, xc)$$

and

$$\beta(a, b) = ax + by.$$

Equivalently,

$$\alpha = \begin{pmatrix} -y \\ x \end{pmatrix}, \quad \beta = \begin{pmatrix} x & y \end{pmatrix}.$$

Step 2: Check that the composition is zero. For $c \in A$,

$$\beta(\alpha(c)) = \beta(-yc, xc) = (-yc)x + (xc)y = -xyc + xyc = 0.$$

Thus

$$\beta \circ \alpha = 0.$$

Step 3: Identify the cokernel. The image of β is the ideal

$$(x, y) \subseteq A.$$

Therefore

$$\operatorname{coker}(\beta) \cong A/(x, y) \cong k.$$

Step 4: Pass to quasi-coherent sheaves. Sheafifying gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}_k^2} \longrightarrow \mathcal{O}_{\mathbb{A}_k^2}^{\oplus 2} \longrightarrow \mathcal{O}_{\mathbb{A}_k^2} \longrightarrow i_* \mathcal{O}_Z \longrightarrow 0.$$

Explicitly, the maps are

$$\mathcal{O}_{\mathbb{A}_k^2} \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} \mathcal{O}_{\mathbb{A}_k^2}^{\oplus 2} \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} \mathcal{O}_{\mathbb{A}_k^2}.$$

Step 5: Take global sections. Since $\mathbb{A}_k^2$ is affine, taking global sections gives back the original exact module sequence:

$$0 \longrightarrow k[x, y] \longrightarrow k[x, y]^{\oplus 2} \longrightarrow k[x, y] \longrightarrow k \longrightarrow 0.$$

Conclusion.

The structure sheaf of the origin in $\mathbb{A}_k^2$ is resolved by finite free quasi-coherent sheaves.

This is an elementary example of a locally free resolution of a closed subscheme.

Example 5: Kernels and cokernels of quasi-coherent morphisms

Let

$$A = k[x], \quad X = \mathbb{A}_k^1,$$

and consider the A -linear map

$$\mu_x : A \longrightarrow A, \quad f \longmapsto xf.$$

Step 1: Compute the kernel module. Since $k[x]$ is a domain,

$$\ker(\mu_x) = 0.$$

Step 2: Compute the cokernel module. The image is

$$xA = (x),$$

so

$$\operatorname{coker}(\mu_x) = A/(x) \cong k.$$

Thus

$$0 \longrightarrow A \xrightarrow{\mu_x} A \longrightarrow k \longrightarrow 0$$

is exact.

Step 3: Sheafify. The induced morphism of quasi-coherent sheaves is

$$\widetilde{\mu}_x : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \quad s \longmapsto xs.$$

Its kernel and cokernel satisfy

$$\ker(\widetilde{\mu}_x) \cong \widetilde{\ker(\mu_x)} = 0,$$

and

$$\operatorname{coker}(\widetilde{\mu}_x) \cong \widetilde{\operatorname{coker}(\mu_x)} \cong \widetilde{k}.$$

The sheaf $\widetilde{k}$ is the structure sheaf of the origin, pushed forward to $\mathbb{A}_k^1$.

Conclusion.

$$\boxed{\ker(\widetilde{\varphi}) \cong \widetilde{\ker(\varphi)}, \quad \operatorname{coker}(\widetilde{\varphi}) \cong \widetilde{\operatorname{coker}(\varphi)}}$$

for module maps on an affine scheme.

Thus kernels, images, and cokernels of morphisms of quasi-coherent sheaves may be computed directly from modules.

Example 6: Localization of a quotient sheaf

Let

$$A = k[x], \quad M = A/(x^2), \quad X = \operatorname{Spec}(A) = \mathbb{A}_k^1.$$

The sheaf

$$\widetilde{M}$$

is supported at the origin and represents the double point

$$V(x^2) \subseteq \mathbb{A}_k^1.$$

We examine its restriction to the open subset

$$D(x) = \mathbb{A}_k^1 \setminus \{0\}.$$

Step 1: Localize the module. By the affine correspondence,

$$\Gamma(D(x), \widetilde{M}) \cong M_x.$$

But

$$M_x = (A/(x^2))_x.$$

Since x becomes invertible in A_x , the relation

$$x^2 = 0$$

implies

$$1 = 0$$

after localization. Therefore

$$M_x = 0.$$

Step 2: Interpret geometrically. The sheaf $\widetilde{M}$ has no sections and no support away from the origin:

$$\widetilde{M}|_{D(x)} = 0.$$

This agrees with the geometry: the double point is supported entirely at $x = 0$.

Conclusion.

$$\Gamma\left(D(x), \widetilde{A/(x^2)}\right) = 0,$$

because localization detects that the closed subscheme $V(x^2)$ does not meet the open subset $D(x)$.

Example 7: Why affineness matters

Exactness of global sections is special to quasi-coherent sheaves on affine schemes. It fails on projective schemes.

Let

$$X = \mathbb{P}_k^1,$$

and let $p \in X$ be a closed point. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-1) \longrightarrow k_p \longrightarrow 0.$$

Step 1: Take global sections. We obtain

$$0 \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(-2)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(-1)) \longrightarrow H^0(\mathbb{P}^1, k_p).$$

Step 2: Substitute the known groups. We have

$$H^0(\mathbb{P}^1, \mathcal{O}(-2)) = 0, \quad H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0,$$

while

$$H^0(\mathbb{P}^1, k_p) \cong k.$$

Therefore the sequence of global sections begins

$$0 \longrightarrow 0 \longrightarrow 0 \longrightarrow k.$$

The final map is not surjective.

Step 3: Locate the obstruction. The long exact sequence continues with

$$k \longrightarrow H^1(\mathbb{P}^1, \mathcal{O}(-2)).$$

In fact,

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k,$$

and the connecting map is an isomorphism.

Conclusion.

$\Gamma(X, -)$ is exact on affine quasi-coherent sheaves, but not generally on projective schemes.

The difference is measured by higher cohomology.

9.6 Exact sequences associated with closed subschemes

Let

$$X = \operatorname{Spec}(A)$$

and let

$$I \subseteq A$$

be an ideal. The closed subscheme determined by I is

$$Z = V(I) = \operatorname{Spec}(A/I).$$

There is an exact module sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0.$$

After passing to quasi-coherent sheaves, one obtains

$$0 \longrightarrow \tilde{I} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Here

$$\tilde{I} = \mathcal{I}_{Z/X}$$

is the ideal sheaf of Z in X .

Thus:

$$\begin{array}{ccc} I \subseteq A & \longleftrightarrow & \mathcal{I}_{Z/X} \subseteq \mathcal{O}_X, \\ A/I & \longleftrightarrow & i_*\mathcal{O}_Z. \end{array}$$

This correspondence explains why closed subschemes of affine schemes are described completely by ideals and quotient rings.

9.7 Finite modules and coherent sheaves

Suppose now that A is Noetherian.

A quasi-coherent sheaf

$$\mathcal{F} \cong \tilde{M}$$

on

$$X = \operatorname{Spec}(A)$$

is coherent if and only if M is a finitely generated A -module.

Because A is Noetherian, kernels and cokernels of homomorphisms between finitely generated A -modules are again finitely generated.

Therefore, if

$$\mathcal{F} \longrightarrow \mathcal{G}$$

is a morphism of coherent sheaves on an affine Noetherian scheme, then

$$\ker(\mathcal{F} \rightarrow \mathcal{G}), \quad \operatorname{im}(\mathcal{F} \rightarrow \mathcal{G}), \quad \operatorname{coker}(\mathcal{F} \rightarrow \mathcal{G})$$

are coherent.

On affine Noetherian schemes, coherent sheaves behave like finite modules.

Example: Coherence of the structure sheaf of a parabola

Let

$$Y = \mathbb{A}_k^2 = \operatorname{Spec}(k[x, y]),$$

and let

$$Z = V(y - x^2) \subseteq Y.$$

The coordinate ring

$$k[x, y]/(y - x^2)$$

is generated by the class of 1 as a module over $k[x, y]$. Therefore it is a finite $k[x, y]$ -module.

Hence

$$i_*\mathcal{O}_Z \cong \widetilde{k[x, y]/(y - x^2)}$$

is a coherent sheaf on Y .

Moreover, the ideal

$$(y - x^2)$$

is finitely generated, so the exact sequence

$$0 \longrightarrow \mathcal{I}_{Z/Y} \longrightarrow \mathcal{O}_Y \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0$$

is an exact sequence of coherent sheaves.

9.8 Comparison table

9.9 A practical workflow for affine quasi-coherent calculations

When a problem involves quasi-coherent sheaves on an affine scheme, the following workflow is useful.

Step 1: Identify the affine coordinate ring

Write

$$X = \operatorname{Spec}(A).$$

This identifies the ring over which all corresponding modules are defined.

Geometric object or process	Affine module description	Associated sheaf construction	Main conclusion
Affine scheme $X = \text{Spec}(A)$	A -modules M	$M \mapsto \widetilde{M}$	$A\text{-Mod} \simeq \text{QCoh}(X)$
Distinguished open subset $D(f)$	Localization M_f	$\Gamma(D(f), \widetilde{M}) \cong M_f$	Restriction of sheaves is computed by localization
Closed subscheme $Z = V(I)$	$0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$	$0 \rightarrow \mathcal{I}_{Z/X} \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_Z \rightarrow 0$	Ideal sheaves are the geometric form of ideals
Reduced origin in $\mathbb{A}^1$	$A/(x) \cong k$	$\mathcal{O}_{\{0\}} \cong \widetilde{A/(x)}$	A quotient by (x) gives a reduced point
Double point in $\mathbb{A}^1$	$A/(x^2)$	$\mathcal{O}_{V(x^2)} \cong \widetilde{A/(x^2)}$	Nilpotents record infinitesimal thickness
Origin in $\mathbb{A}^2$	Koszul resolution of $A/(x, y)$	Resolution by finite free sheaves	Closed subschemes may have locally free resolutions
Global sections on affine X	Exact sequences of A -modules	Exact sequences in $\text{QCoh}(X)$	$\Gamma(X, -)$ is exact on quasi-coherent sheaves
Projective contrast $X = \mathbb{P}^1$	No direct affine module equivalence globally	$0 \rightarrow \mathcal{O}(-2) \rightarrow \mathcal{O}(-1) \rightarrow k_p \rightarrow 0$	Global sections need not preserve surjectivity

Table 2: Exact sequences, quasi-coherent sheaves, and global sections in affine geometry.

Step 2: Replace quasi-coherent sheaves by modules

If

$$\mathcal{F}$$

is quasi-coherent, write

$$\mathcal{F} \cong \widetilde{M}, \quad M = \Gamma(X, \mathcal{F}).$$

Likewise, replace morphisms of quasi-coherent sheaves by module homomorphisms.

Step 3: Compute kernels, images, quotients, and exactness algebraically

For a map

$$\varphi : M \longrightarrow N,$$

compute

$$\ker(\varphi), \quad \text{im}(\varphi), \quad \text{coker}(\varphi)$$

as ordinary modules.

Then pass back to sheaves:

$$\ker(\widetilde{\varphi}) \cong \widetilde{\ker(\varphi)},$$

$$\text{im}(\widetilde{\varphi}) \cong \widetilde{\text{im}(\varphi)},$$

and

$$\text{coker}(\widetilde{\varphi}) \cong \widetilde{\text{coker}(\varphi)}.$$

Step 4: Use localization on distinguished opens

If one needs local information on

$$D(f) \subseteq X,$$

compute

$$M_f.$$

Then

$$\Gamma(D(f), \widetilde{M}) \cong M_f.$$

This determines where the sheaf vanishes, where generators become invertible, and how the corresponding geometric object behaves locally.

Step 5: Recover global sections without higher obstruction

Since X is affine and the sheaves are quasi-coherent, take global sections directly:

$$\Gamma(X, \widetilde{M}) \cong M.$$

Moreover, any short exact sequence of quasi-coherent sheaves remains exact after applying $\Gamma(X, -)$.

Thus no H^1 -obstruction appears in the affine quasi-coherent setting.

Step 6: Compare with the non-affine case

If the same reasoning is attempted on a projective scheme, global sections may cease to be exact. The missing surjectivity is then measured by cohomology groups such as

$$H^1(X, \mathcal{F}).$$

This comparison clarifies why affine open subsets are the fundamental local building blocks of schemes.

9.10 Final conceptual summary

The theory of quasi-coherent sheaves on affine schemes is powerful because it turns geometric constructions into module calculations.

For

$$X = \operatorname{Spec}(A),$$

one has the equivalence

$$A\text{-}\mathbf{Mod} \simeq \operatorname{QCoh}(X), \quad M \longleftrightarrow \widetilde{M}.$$

Under this correspondence:

$$\Gamma(X, \widetilde{M}) \cong M,$$

$$\Gamma(D(f), \widetilde{M}) \cong M_f,$$

and exact sequences of modules correspond to exact sequences of quasi-coherent sheaves.

A closed subscheme

$$Z = V(I) \subseteq X$$

is represented by the quotient module

$$A/I$$

and the ideal-sheaf sequence

$$0 \longrightarrow \widetilde{I} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Reduced points, thickened points, affine curves, and finite resolutions of closed subschemes are all instances of this same mechanism.

Finally, affineness eliminates higher global obstructions for quasi-coherent sheaves:

$$H^i(X, \mathcal{F}) = 0 \quad \text{for every } i > 0.$$

Therefore:

modules over $A \iff$ quasi-coherent sheaves on $\text{Spec}(A)$,
 exact module sequences $\iff$ exact quasi-coherent sheaf sequences,
 affine global sections $\iff$ exact algebraic recovery of modules.

10 Cohomology and Acyclic Resolutions

Throughout this section, let X be a topological space or a scheme, and let $\mathcal{F}$ be a sheaf of abelian groups or an $\mathcal{O}_X$ -module.

The central reason cohomology appears is that global sections do not always preserve surjectivity. A section may exist locally everywhere and yet fail to arise from one global section.

The guiding principle is:

Cohomology measures the obstruction to passing from local data to global data.

In algebraic geometry, this becomes especially important for sheaves such as

$$\mathcal{O}_X, \quad \mathcal{O}_X(d), \quad \widetilde{M}, \quad \mathcal{I}_{Z/X},$$

and for exact sequences involving line bundles, ideal sheaves, and closed subschemes.

10.1 Why cohomology appears

Let

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be a short exact sequence of sheaves.

Applying global sections gives

$$0 \longrightarrow \Gamma(X, \mathcal{F}') \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{F}'').$$

This sequence is always exact at the first two positions. However, the map

$$\Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{F}'')$$

need not be surjective.

A section of $\mathcal{F}''$ may lift locally to sections of $\mathcal{F}$, but these local lifts may fail to agree on overlaps. Cohomology completes the sequence into the long exact sequence

$$\begin{aligned} 0 &\longrightarrow H^0(X, \mathcal{F}') \longrightarrow H^0(X, \mathcal{F}) \longrightarrow H^0(X, \mathcal{F}'') \\ &\longrightarrow H^1(X, \mathcal{F}') \longrightarrow H^1(X, \mathcal{F}) \longrightarrow H^1(X, \mathcal{F}'') \\ &\longrightarrow H^2(X, \mathcal{F}') \longrightarrow \cdots, \end{aligned}$$

where

$$H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F}).$$

The connecting morphism

$$\delta : H^0(X, \mathcal{F}'') \longrightarrow H^1(X, \mathcal{F}')$$

sends a global section of $\mathcal{F}''$ to the obstruction to lifting it globally through $\mathcal{F}$.

Thus:

H^1 detects failures of global lifting.

Higher groups $H^2, H^3, \dots$ record higher-order gluing obstructions.

10.2 Definition of sheaf cohomology

The global-section functor is

$$\Gamma(X, -) : \mathbf{Sh}(X) \longrightarrow \mathbf{Ab}.$$

It is left exact, but it is not generally exact. The cohomology groups of a sheaf $\mathcal{F}$ are the right derived functors of global sections:

$$H^i(X, \mathcal{F}) = R^i\Gamma(X, \mathcal{F}).$$

Thus

$$H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F}),$$

while

$$H^1(X, \mathcal{F}), \quad H^2(X, \mathcal{F}), \quad \dots$$

measure the failure of $\Gamma(X, -)$ to preserve exactness.

For a scheme X and a quasi-coherent sheaf $\mathcal{F}$, these groups connect local module theory with global geometric phenomena.

10.3 Resolutions and acyclic sheaves

Injective resolutions

An *injective resolution* of a sheaf $\mathcal{F}$ is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \mathcal{I}^2 \longrightarrow \cdots,$$

where each $\mathcal{I}^j$ is injective.

After applying global sections, one obtains a cochain complex

$$0 \longrightarrow \Gamma(X, \mathcal{I}^0) \longrightarrow \Gamma(X, \mathcal{I}^1) \longrightarrow \Gamma(X, \mathcal{I}^2) \longrightarrow \cdots.$$

The cohomology of this complex computes the sheaf cohomology:

$$H^i(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{I}^\bullet)).$$

Injective resolutions are fundamental theoretically, but they are often not the most convenient objects for explicit calculations.

Acyclic sheaves

Definition 10.3.1 (Acyclic sheaf). A sheaf $\mathcal{A}$ is called *acyclic for global sections* if

$$H^i(X, \mathcal{A}) = 0 \quad \text{for every } i > 0.$$

Important examples include:

- injective sheaves;
- flasque sheaves;
- fine sheaves on suitable topological spaces;
- quasi-coherent sheaves on affine schemes.

The last example is particularly important in algebraic geometry.

Theorem 10.1 (Affine vanishing). *Let*

$$X = \operatorname{Spec}(A)$$

be an affine scheme and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. Then

$$H^i(X, \mathcal{F}) = 0 \quad \text{for every } i > 0.$$

Thus quasi-coherent sheaves on affine schemes are acyclic for global sections.

Acyclic resolutions

Definition 10.3.2 (Acyclic resolution). An *acyclic resolution* of $\mathcal{F}$ is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{A}^0 \longrightarrow \mathcal{A}^1 \longrightarrow \mathcal{A}^2 \longrightarrow \cdots$$

such that each $\mathcal{A}^j$ is acyclic for global sections.

If $\mathcal{A}^\bullet$ is an acyclic resolution of $\mathcal{F}$, then

$$H^i(X, \mathcal{F}) \cong H^i(\Gamma(X, \mathcal{A}^\bullet)).$$

This is often far easier than constructing an injective resolution directly.

10.4 Flasque resolutions and the Godement construction

Definition 10.4.1 (Flasque sheaf). A sheaf $\mathcal{A}$ is *flasque* if every restriction map

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(V)$$

is surjective whenever $V \subseteq U$ are open subsets.

Flasque sheaves are acyclic:

$$\mathcal{A} \text{ flasque} \implies H^i(X, \mathcal{A}) = 0 \text{ for every } i > 0.$$

A general way of producing a flasque resolution is the Godement construction. For a sheaf $\mathcal{F}$, define

$$\mathcal{C}^0(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x.$$

Thus a section of $\mathcal{C}^0(\mathcal{F})$ over U is an arbitrary family of germs, one germ at each point of U .

There is a natural morphism

$$\mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{F})$$

sending a section to its family of germs. This morphism is injective by the locality axiom of a sheaf.

The sheaf $\mathcal{C}^0(\mathcal{F})$ is flasque, because restricting a family of germs merely forgets some components. Let

$$\mathcal{F}^1 = \operatorname{coker}(\mathcal{F} \rightarrow \mathcal{C}^0(\mathcal{F})),$$

and define

$$\mathcal{C}^1(\mathcal{F}) = \mathcal{C}^0(\mathcal{F}^1).$$

Continuing inductively gives a flasque resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{F}) \longrightarrow \mathcal{C}^2(\mathcal{F}) \longrightarrow \cdots.$$

This proves abstractly that every sheaf admits an acyclic resolution. For concrete computations on schemes, however, affine covers and Čech complexes are usually more effective.

10.5 Čech cohomology from an affine cover

Let

$$X = U_0 \cup U_1$$

be an open cover, and let $\mathcal{F}$ be a sheaf. The Čech complex for this two-open cover is

$$0 \longrightarrow \Gamma(U_0, \mathcal{F}) \oplus \Gamma(U_1, \mathcal{F}) \xrightarrow{\delta} \Gamma(U_0 \cap U_1, \mathcal{F}) \longrightarrow 0,$$

where

$$\delta(s_0, s_1) = s_1|_{U_0 \cap U_1} - s_0|_{U_0 \cap U_1}.$$

Then

$$\check{H}^0(\mathcal{U}, \mathcal{F}) = \ker(\delta),$$

and

$$\check{H}^1(\mathcal{U}, \mathcal{F}) = \operatorname{coker}(\delta).$$

If the cover is acyclic for $\mathcal{F}$, then Čech cohomology agrees with sheaf cohomology:

$$\check{H}^i(\mathcal{U}, \mathcal{F}) \cong H^i(X, \mathcal{F}).$$

For quasi-coherent sheaves on separated schemes covered by affine opens whose finite intersections are affine, this gives an explicit computational method.

10.6 Step-by-step examples

Example 1: Cohomology on an affine scheme

Let

$$X = \operatorname{Spec}(A)$$

and let M be an A -module. The associated quasi-coherent sheaf is denoted

$$\widetilde{M}.$$

The fundamental affine correspondence is

$$\Gamma(X, \widetilde{M}) \cong M.$$

Moreover,

$$H^i(X, \widetilde{M}) = 0 \quad \text{for every } i > 0.$$

We now apply this to the affine line

$$X = \mathbb{A}_k^1 = \operatorname{Spec}(k[t]).$$

Its structure sheaf is

$$\mathcal{O}_X = \widetilde{k[t]}.$$

Step 1: Compute global sections. By the affine correspondence,

$$\Gamma(\mathbb{A}_k^1, \mathcal{O}_X) \cong k[t].$$

Therefore

$$H^0(\mathbb{A}_k^1, \mathcal{O}_X) = k[t].$$

Step 2: Compute higher cohomology. Since $\mathbb{A}_k^1$ is affine and $\mathcal{O}_X$ is quasi-coherent,

$$H^i(\mathbb{A}_k^1, \mathcal{O}_X) = 0 \quad \text{for every } i > 0.$$

Conclusion.

$$\boxed{H^0(\mathbb{A}_k^1, \mathcal{O}_X) = k[t], \quad H^i(\mathbb{A}_k^1, \mathcal{O}_X) = 0 \text{ for } i > 0.}$$

On an affine scheme, quasi-coherent sheaves have no higher global obstruction. Their global behaviour is controlled directly by modules.

Example 2: Computing $H^1(\mathbb{P}^1, \mathcal{O}(-2))$

Consider the projective line

$$X = \mathbb{P}_k^1.$$

Unlike affine space, projective space may have nonzero higher cohomology. We compute

$$H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2)).$$

Step 1: Choose the standard affine cover. Let

$$U_0 = D_+(s), \quad U_1 = D_+(t).$$

Then

$$U_0 \cong \mathbb{A}_k^1, \quad U_1 \cong \mathbb{A}_k^1.$$

On U_0 , introduce the coordinate

$$u = \frac{t}{s}.$$

On U_1 , the corresponding coordinate is

$$u^{-1} = \frac{s}{t}.$$

Their intersection is

$$U_0 \cap U_1 \cong \text{Spec}(k[u, u^{-1}]).$$

Step 2: Trivialize the line bundle on each chart. Choose a frame e_0 for $\mathcal{O}(-2)$ over U_0 , and a frame e_1 over U_1 . On the overlap, they satisfy

$$e_1 = u^{-2}e_0.$$

Therefore

$$\Gamma(U_0, \mathcal{O}(-2)) = k[u]e_0,$$

while sections over U_1 , expressed using the e_0 -frame, are

$$u^{-2}k[u^{-1}]e_0.$$

Over the intersection,

$$\Gamma(U_0 \cap U_1, \mathcal{O}(-2)) = k[u, u^{-1}]e_0.$$

Step 3: Write the Čech complex. The Čech complex is

$$0 \longrightarrow k[u]e_0 \oplus u^{-2}k[u^{-1}]e_0 \longrightarrow k[u, u^{-1}]e_0 \longrightarrow 0.$$

Hence

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong \frac{k[u, u^{-1}]}{k[u] + u^{-2}k[u^{-1}]}.$$

Step 4: Determine which Laurent monomials survive. The vector space $k[u, u^{-1}]$ is spanned by

$$\dots, u^{-3}, u^{-2}, u^{-1}, 1, u, u^2, \dots$$

The subspace $k[u]$ contains

$$1, u, u^2, \dots,$$

and the subspace $u^{-2}k[u^{-1}]$ contains

$$u^{-2}, u^{-3}, u^{-4}, \dots$$

Every Laurent monomial is therefore removed except

$$u^{-1}.$$

Thus

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k \cdot [u^{-1}].$$

Conclusion.

$$\boxed{H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k.}$$

This is one of the simplest examples of a nontrivial cohomological obstruction on projective space.

Example 3: Cohomology of $\mathcal{O}_{\mathbb{P}^1}(n)$

The previous calculation generalizes. Using the same standard affine cover of $\mathbb{P}^1$, one obtains

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) \cong \frac{k[u, u^{-1}]}{k[u] + u^n k[u^{-1}]}.$$

We now analyze the possibilities for n .

Case 1: $n \geq 0$. For $n \geq 0$, the spaces

$$k[u] \quad \text{and} \quad u^n k[u^{-1}]$$

together contain all Laurent monomials. Hence

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0 \quad \text{for } n \geq 0.$$

At the same time,

$$H^0(\mathbb{P}^1, \mathcal{O}(n))$$

has basis

$$s^n, s^{n-1}t, \dots, st^{n-1}, t^n.$$

Therefore

$$\dim_k H^0(\mathbb{P}^1, \mathcal{O}(n)) = n + 1.$$

Case 2: $n = -1$. Here

$$H^1(\mathbb{P}^1, \mathcal{O}(-1)) \cong \frac{k[u, u^{-1}]}{k[u] + u^{-1}k[u^{-1}]}.$$

The first subspace contains all nonnegative powers of u , while the second contains all negative powers. Hence

$$H^1(\mathbb{P}^1, \mathcal{O}(-1)) = 0.$$

Also,

$$H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0.$$

Therefore

$$\boxed{H^0(\mathbb{P}^1, \mathcal{O}(-1)) = H^1(\mathbb{P}^1, \mathcal{O}(-1)) = 0.}$$

Case 3: $n = -2$. As computed above,

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k,$$

generated by the class of

$$u^{-1}.$$

Case 4: $n = -3$. Now

$$H^1(\mathbb{P}^1, \mathcal{O}(-3)) \cong \frac{k[u, u^{-1}]}{k[u] + u^{-3}k[u^{-1}]}.$$

The surviving monomials are

$$u^{-1}, \quad u^{-2}.$$

Therefore

$$H^1(\mathbb{P}^1, \mathcal{O}(-3)) \cong k^2.$$

General result. For every integer n ,

$$\dim_k H^0(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0, \end{cases}$$

and

$$\dim_k H^1(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n \leq -2. \end{cases}$$

Moreover,

$$H^i(\mathbb{P}^1, \mathcal{O}(n)) = 0 \quad \text{for every } i \geq 2.$$

Example 4: Serre duality on $\mathbb{P}^1$

The canonical line bundle of the projective line is

$$\omega_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2).$$

Serre duality states that

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) \cong H^0(\mathbb{P}^1, \mathcal{O}(-n-2))^\vee.$$

We verify this in two explicit cases.

Step 1: Take $n = -2$. Serre duality gives

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong H^0(\mathbb{P}^1, \mathcal{O})^\vee.$$

Since

$$H^0(\mathbb{P}^1, \mathcal{O}) \cong k,$$

we obtain

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k^\vee \cong k.$$

This agrees with the Čech calculation.

Step 2: Take $n = -3$. Serre duality gives

$$H^1(\mathbb{P}^1, \mathcal{O}(-3)) \cong H^0(\mathbb{P}^1, \mathcal{O}(1))^\vee.$$

Since

$$H^0(\mathbb{P}^1, \mathcal{O}(1)) = \langle s, t \rangle \cong k^2,$$

it follows that

$$H^1(\mathbb{P}^1, \mathcal{O}(-3)) \cong (k^2)^\vee \cong k^2.$$

Again, this agrees with the Čech calculation.

Interpretation. Positive line bundles have many global sections. Sufficiently negative line bundles have no global sections, but their negativity appears as nontrivial first cohomology.

On $\mathbb{P}^1$, Serre duality exchanges

global sections of positive bundles $\longleftrightarrow$ first cohomology of sufficiently negative bundles.
--

Example 5: A connecting morphism as an obstruction

Let

$$p \in \mathbb{P}_k^1$$

be a closed point. The ideal sheaf of p is

$$\mathcal{I}_p \cong \mathcal{O}_{\mathbb{P}^1}(-1).$$

Therefore there is an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-1) \longrightarrow \mathcal{O}_{\mathbb{P}^1} \longrightarrow k_p \longrightarrow 0,$$

where k_p is the skyscraper sheaf supported at p .

Tensor this sequence by $\mathcal{O}_{\mathbb{P}^1}(-1)$. After choosing an identification of the one-dimensional fiber at p with k , we obtain

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-1) \longrightarrow k_p \longrightarrow 0.$$

Step 1: Apply global sections. The resulting long exact sequence begins

$$\begin{aligned} 0 &\longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(-2)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(-1)) \\ &\longrightarrow H^0(\mathbb{P}^1, k_p) \xrightarrow{\delta} H^1(\mathbb{P}^1, \mathcal{O}(-2)) \\ &\longrightarrow H^1(\mathbb{P}^1, \mathcal{O}(-1)) \longrightarrow 0. \end{aligned}$$

Step 2: Insert the known groups. We know that

$$\begin{aligned} H^0(\mathbb{P}^1, \mathcal{O}(-2)) &= 0, & H^0(\mathbb{P}^1, \mathcal{O}(-1)) &= 0, \\ H^0(\mathbb{P}^1, k_p) &\cong k, & H^1(\mathbb{P}^1, \mathcal{O}(-2)) &\cong k, \end{aligned}$$

and

$$H^1(\mathbb{P}^1, \mathcal{O}(-1)) = 0.$$

Thus the long exact sequence becomes

$$0 \longrightarrow 0 \longrightarrow 0 \longrightarrow k \xrightarrow{\delta} k \longrightarrow 0.$$

Hence

$$\delta : k \longrightarrow k$$

is an isomorphism.

Step 3: Interpret the obstruction. A nonzero section of k_p is a nonzero value in the fiber at the point p . It cannot arise from a global section of $\mathcal{O}(-1)$, because

$$H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0.$$

Locally, such a lift exists; globally, it does not. The nonzero obstruction is precisely its image under

$$\delta : H^0(k_p) \longrightarrow H^1(\mathcal{O}(-2)).$$

Therefore

$H^1(\mathbb{P}^1, \mathcal{O}(-2))$ records a concrete failure of global lifting.

Example 6: The Čech complex as an acyclic-resolution computation

Let

$$X = \mathbb{P}_k^1$$

and take the standard affine cover

$$X = U_0 \cup U_1, \quad U_0 = D_+(s), \quad U_1 = D_+(t).$$

Let

$$\mathcal{F} = \mathcal{O}_{\mathbb{P}^1}(-2).$$

Since

$$U_0, \quad U_1, \quad U_0 \cap U_1$$

are affine, and since $\mathcal{F}$ is quasi-coherent, every restriction of $\mathcal{F}$ to these affine opens has vanishing higher cohomology:

$$H^i(U_0, \mathcal{F}) = 0, \quad H^i(U_1, \mathcal{F}) = 0, \quad H^i(U_0 \cap U_1, \mathcal{F}) = 0$$

for every $i > 0$.

Thus this cover is acyclic for $\mathcal{F}$, and its Čech complex computes the sheaf cohomology.

Step 1: Write the complex of sections. Using

$$u = \frac{t}{s},$$

the Čech complex becomes

$$0 \longrightarrow k[u] \oplus u^{-2}k[u^{-1}] \xrightarrow{\delta} k[u, u^{-1}] \longrightarrow 0.$$

Step 2: Compute degree-zero cohomology. The kernel of δ consists of pairs of local sections that agree on the overlap. For $\mathcal{O}(-2)$, there is no nonzero global section. Hence

$$H^0(\mathbb{P}^1, \mathcal{O}(-2)) = 0.$$

Step 3: Compute degree-one cohomology. The cokernel is

$$\frac{k[u, u^{-1}]}{k[u] + u^{-2}k[u^{-1}]}.$$

Only the Laurent monomial

$$u^{-1}$$

survives. Therefore

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k.$$

Step 4: Interpret the calculation. The affine pieces are cohomologically simple: higher cohomology vanishes on each piece. All nontrivial global information appears in the failure of local sections to agree on the overlap.

Thus

Affine pieces are locally simple, while overlaps contain the global obstruction.

This is the computational meaning of using an acyclic resolution.

Example 7: Cohomology and a closed immersion

Let

$$i : Z \hookrightarrow X$$

be a closed immersion with ideal sheaf

$$\mathcal{I}_{Z/X}.$$

There is an exact sequence

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Applying global sections gives

$$\begin{aligned} 0 \longrightarrow H^0(X, \mathcal{I}_{Z/X}) &\longrightarrow H^0(X, \mathcal{O}_X) \longrightarrow H^0(Z, \mathcal{O}_Z) \\ &\longrightarrow H^1(X, \mathcal{I}_{Z/X}) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow \cdots \end{aligned}$$

This sequence determines whether functions on Z extend to functions on X .

Take

$$Z = \{p\} \subseteq \mathbb{P}^1.$$

Then

$$\mathcal{I}_p \cong \mathcal{O}(-1),$$

so

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \mathcal{O} \longrightarrow k_p \longrightarrow 0.$$

Taking cohomology gives

$$0 \longrightarrow H^0(\mathcal{O}(-1)) \longrightarrow H^0(\mathcal{O}) \longrightarrow H^0(k_p) \longrightarrow H^1(\mathcal{O}(-1)) \longrightarrow 0.$$

Substituting the known groups yields

$$0 \longrightarrow 0 \longrightarrow k \longrightarrow k \longrightarrow 0 \longrightarrow 0.$$

The map

$$k \longrightarrow k$$

is evaluation of a constant function at p , hence it is an isomorphism. Therefore every regular function on the reduced point extends to a global regular function on $\mathbb{P}^1$.

By contrast, after twisting by $\mathcal{O}(-1)$, the corresponding lifting fails, and the obstruction appears in

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)).$$

10.7 Relation to linear systems and embeddings

Cohomology is deeply connected with the linear systems used to construct projective morphisms.

Let $\mathcal{L}$ be a line bundle on X , and let $p \in X$. The exact sequence

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{I}_p \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_p \longrightarrow 0$$

gives a segment of the long exact sequence

$$H^0(X, \mathcal{L}) \longrightarrow \mathcal{L}|_p \longrightarrow H^1(X, \mathcal{L} \otimes \mathcal{I}_p).$$

If

$$H^1(X, \mathcal{L} \otimes \mathcal{I}_p) = 0,$$

then every value in the fiber $\mathcal{L}|_p$ is attained by a global section. This is the cohomological mechanism behind global generation.

Similarly, for two points $p, q \in X$, the exact sequence

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{I}_{\{p,q\}} \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_p \oplus \mathcal{L}|_q \longrightarrow 0$$

shows that vanishing of

$$H^1(X, \mathcal{L} \otimes \mathcal{I}_{\{p,q\}})$$

helps guarantee that global sections separate points.

For tangent vectors, one uses the first-order thickening $2p$. The exact sequence

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{I}_{2p} \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_{2p} \longrightarrow 0$$

shows that vanishing of

$$H^1(X, \mathcal{L} \otimes \mathcal{I}_{2p})$$

allows global sections to control first-order infinitesimal data at p .

Thus:

vanishing of suitable H^1 -groups
$\implies$
global generation and separation properties
$\implies$
projective embeddings derived from linear systems.

10.8 Comparison table

Situation	Exact sequence or complex	Cohomological conclusion	Geometric meaning
Affine scheme $X = \text{Spec}(A)$	Quasi-coherent sheaf $\widetilde{M}$ from an A -module M	$H^i(X, \widetilde{M}) = 0$ for $i > 0$	Affine geometry is controlled by modules
Line bundle $\mathcal{O}_{\mathbb{P}^1}(-2)$	Čech complex for $U_0 \cup U_1$	$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$	A global obstruction appears on projective space
Point $p \subseteq \mathbb{P}^1$	$0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{O} \rightarrow k_p \rightarrow 0$	$H^0(\mathcal{O}) \rightarrow H^0(k_p)$ is surjective	Regular functions on the reduced point extend globally
Twisted point sequence	$0 \rightarrow \mathcal{O}(-2) \rightarrow \mathcal{O}(-1) \rightarrow k_p \rightarrow 0$	$H^0(k_p) \cong H^1(\mathcal{O}(-2))$	A fiber value fails to lift globally
Linear system from a line bundle	Sequences involving $\mathcal{L} \otimes \mathcal{I}_p$ and $\mathcal{L} \otimes \mathcal{I}_{2p}$	Vanishing of suitable H^1 -groups gives separation	Cohomology helps prove closed immersion criteria

Table 3: The role of cohomology and acyclic resolutions in geometric constructions.

10.9 A practical workflow for cohomology computations

When asked to compute or interpret a cohomology group, the following workflow is useful.

Step 1: Determine whether the scheme is affine

If

$$X = \operatorname{Spec}(A)$$

and $\mathcal{F}$ is quasi-coherent, immediately use

$$H^i(X, \mathcal{F}) = 0 \quad \text{for every } i > 0.$$

Step 2: Look for a useful affine cover

For projective schemes, choose standard affine opens. For example, on $\mathbb{P}^1$,

$$\mathbb{P}^1 = D_+(s) \cup D_+(t).$$

Compute sections on

$$U_0, \quad U_1, \quad U_0 \cap U_1.$$

Step 3: Write the Čech complex

For a two-open cover, use

$$0 \longrightarrow \Gamma(U_0, \mathcal{F}) \oplus \Gamma(U_1, \mathcal{F}) \longrightarrow \Gamma(U_0 \cap U_1, \mathcal{F}) \longrightarrow 0.$$

Then compute

$$H^0 = \ker(\delta), \quad H^1 = \operatorname{coker}(\delta).$$

Step 4: Use exact sequences whenever a subscheme appears

For a closed immersion

$$Z \hookrightarrow X,$$

use

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z \longrightarrow 0.$$

For a line bundle $\mathcal{L}$, tensor the sequence:

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{I}_{Z/X} \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_Z \longrightarrow 0.$$

Then examine the resulting long exact sequence in cohomology.

Step 5: Interpret the result geometrically

A nonzero cohomology group is not merely an algebraic calculation. It records a geometric obstruction:

Cohomological phenomenon	Geometric meaning
$H^1 \neq 0$	Local lifts may fail to glue globally
$H^1(\mathcal{L} \otimes \mathcal{I}_p) = 0$	Global sections can prescribe values at p
$H^1(\mathcal{L} \otimes \mathcal{I}_{\{p,q\}}) = 0$	Global sections can separate points
$H^1(\mathcal{L} \otimes \mathcal{I}_{2p}) = 0$	Global sections can separate tangent directions

10.10 Final conceptual summary

Sheaf cohomology begins with the simple fact that

$$\Gamma(X, -)$$

is left exact, but it is not always exact.

Acyclic resolutions replace a difficult sheaf by sheaves whose higher cohomology vanishes. On affine schemes, quasi-coherent sheaves are already acyclic. On projective schemes, one often covers the scheme by affine opens and computes with a Čech complex.

The fundamental examples on $\mathbb{P}^1$ are:

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \cong \begin{cases} k^{n+1}, & n \geq 0, \\ 0, & n < 0, \end{cases}$$

and

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) \cong \begin{cases} 0, & n \geq -1, \\ k^{-n-1}, & n \leq -2. \end{cases}$$

In particular,

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k,$$

and this one-dimensional group measures a concrete failure of global lifting.

Finally, cohomology is not separate from embeddings and linear systems. Vanishing of suitable first cohomology groups ensures that sections of line bundles can control values, distinguish points, and detect tangent directions.

Thus:

affine acyclicity $\implies$ local algebraic control, Čech and acyclic resolutions $\implies$ explicit global computations, cohomological vanishing $\implies$ geometric generation and embedding results.
--

11 Extensions and Connections with Other Notions

The three results are not isolated facts. They are connected with several central notions in algebraic geometry: very ample line bundles, projective embeddings, affine schemes, quasi-coherent sheaves, derived functors, and sheaf cohomology.

11.1 Connection with Very Ample Line Bundles

The closed immersion criterion is one of the main tools for studying very ample line bundles.

A line bundle $\mathcal{L}$ on a scheme X is called very ample if there exists a closed immersion

$$X \hookrightarrow \mathbb{P}^n$$

such that

$$\mathcal{L} \cong \mathcal{O}_{\mathbb{P}^n}(1)|_X.$$

Thus very ampleness means that the global sections of $\mathcal{L}$ embed X into projective space. The criterion says that this happens precisely when the sections separate both points and tangent vectors:

$$\boxed{\mathcal{L} \text{ is very ample} \iff \Gamma(X, \mathcal{L}) \text{ separates points and tangent vectors.}}$$

Therefore, very ampleness is a geometric condition expressed in terms of global sections.

11.2 Connection with Projective Embeddings

If X is projective, then one often writes

$$X = \text{Proj}(S)$$

for some graded ring S .

The graded pieces

$$S_d$$

correspond to global sections of twists

$$\mathcal{O}_X(d).$$

Thus the spaces

$$H^0(X, \mathcal{O}_X(d))$$

produce morphisms

$$X \rightarrow \mathbb{P}^N.$$

For sufficiently positive d , these morphisms often become closed immersions. Hence the closed immersion criterion explains how projective varieties are embedded into projective space using homogeneous coordinates.

11.3 Connection with Tangent Spaces

Separating tangent vectors is a condition on the cotangent space

$$\mathfrak{m}_P/\mathfrak{m}_P^2.$$

The Zariski tangent space is its dual:

$$T_P X = \text{Hom}_k(\mathfrak{m}_P/\mathfrak{m}_P^2, k).$$

Thus a morphism separates tangent vectors when its coordinate functions detect all first-order directions at a point.

In this way, the closed immersion criterion is closely connected with:

Zariski tangent spaces,
differentials,
smoothness,

and

the Jacobian criterion.

The basic relationship is:

$$\boxed{\text{sections vanishing at } P \implies \mathfrak{m}_P/\mathfrak{m}_P^2 \implies T_P X.}$$

11.4 Connection with Serre's Theorem on Affine Schemes

The exactness of global sections on affine schemes is a special case of a deeper theorem of Serre.

For a scheme X , the following condition characterizes affineness:

$$X \text{ is affine} \iff H^i(X, \mathcal{F}) = 0 \quad \text{for all } i > 0$$

and for all quasi-coherent sheaves $\mathcal{F}$.

Thus affine schemes are precisely the schemes on which quasi-coherent sheaves have no higher cohomology.

The exercise

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

implies

$$0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0$$

when X is affine and the sheaves are quasi-coherent.

This is the degree-zero manifestation of the vanishing theorem:

$$H^i(X, \mathcal{F}) = 0 \quad (i > 0).$$

11.5 Connection with Čech Cohomology

Sheaf cohomology can also be computed using Čech cohomology.

Given an open cover

$$\mathfrak{U} = \{U_i\}_{i \in I}$$

of X , one forms the Čech complex

$$\check{C}^0(\mathfrak{U}, \mathcal{F}) \rightarrow \check{C}^1(\mathfrak{U}, \mathcal{F}) \rightarrow \check{C}^2(\mathfrak{U}, \mathcal{F}) \rightarrow \cdots$$

Under suitable hypotheses, this computes sheaf cohomology:

$$\check{H}^i(\mathfrak{U}, \mathcal{F}) \cong H^i(X, \mathcal{F}).$$

Thus acyclic resolutions and Čech complexes are two major ways of computing the same objects:

$$H^i(X, \mathcal{F}).$$

Method	Computes
Acyclic resolution	$H^i(X, \mathcal{F})$
Čech complex	$H^i(X, \mathcal{F})$

11.6 Connection with Derived Functors

The global section functor

$$\Gamma(X, -)$$

is left exact, but not exact in general.

Its right derived functors are denoted by

$$R^i\Gamma(X, -).$$

By definition,

$$H^i(X, \mathcal{F}) = R^i\Gamma(X, \mathcal{F}).$$

Therefore sheaf cohomology is the derived functor of global sections.

The acyclic resolution theorem says that if

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{F}^0 \rightarrow \mathcal{F}^1 \rightarrow \dots$$

is a resolution by $\Gamma(X, -)$ -acyclic sheaves, then

$$R^i\Gamma(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{F}^\bullet)).$$

This is one of the fundamental principles of homological algebra.

11.7 Connection with Injective Resolutions

Every sheaf of abelian groups admits an injective resolution:

$$0 \rightarrow \mathcal{F} \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

Injective sheaves are acyclic for the global section functor, so

$$H^i(X, \mathcal{F}) = H^i(\Gamma(X, I^\bullet)).$$

Thus injective resolutions provide the general construction of sheaf cohomology.

Acyclic resolutions are often more convenient than injective resolutions because they may be more explicit, for example:

flasque resolutions,

Godement resolutions,

Čech resolutions,

or

fine resolutions on manifolds.

11.8 Connection with Derived Categories

In modern algebraic geometry, one often works with complexes of sheaves rather than individual sheaves.

The derived global section functor is written

$$R\Gamma(X, -).$$

It is a functor

$$R\Gamma(X, -) : D(X) \rightarrow D(\mathbf{Ab}),$$

where $D(X)$ is the derived category of sheaves on X .

For a sheaf $\mathcal{F}$, its cohomology groups are recovered as

$$H^i(X, \mathcal{F}) = H^i(R\Gamma(X, \mathcal{F})).$$

Thus the acyclic resolution theorem is an early concrete form of derived category theory.

11.9 Connection with Vector Bundles and Grassmannians

The closed immersion criterion for line bundles has a higher-rank analogue.

If $\mathcal{E}$ is a vector bundle on X , then sufficiently many global sections of $\mathcal{E}$ may define a morphism

$$X \rightarrow \mathrm{Gr}(r, n),$$

where $\mathrm{Gr}(r, n)$ is a Grassmannian.

Thus:

Bundle	Sections define maps to	Geometry
Line bundle	$\mathbb{P}^n$	projective embeddings
Vector bundle	$\mathrm{Gr}(r, n)$	Grassmannian embeddings

So projective embeddings are the rank-one case of a broader theory of maps defined by vector bundles.

11.10 Connection with Serre Duality

Once one studies sheaf cohomology, a major next theorem is Serre duality.

If X is a smooth projective variety of dimension n , then cohomology groups of a sheaf $\mathcal{F}$ are related to cohomology groups of its dual:

$$H^i(X, \mathcal{F}) \quad \text{and} \quad H^{n-i}(X, \mathcal{F}^\vee \otimes \omega_X).$$

In its simplest form, Serre duality says

$$H^i(X, \mathcal{F})^\vee \cong H^{n-i}(X, \mathcal{F}^\vee \otimes \omega_X),$$

under suitable hypotheses.

Thus the cohomological ideas from acyclic resolutions lead naturally to:

Riemann–Roch,

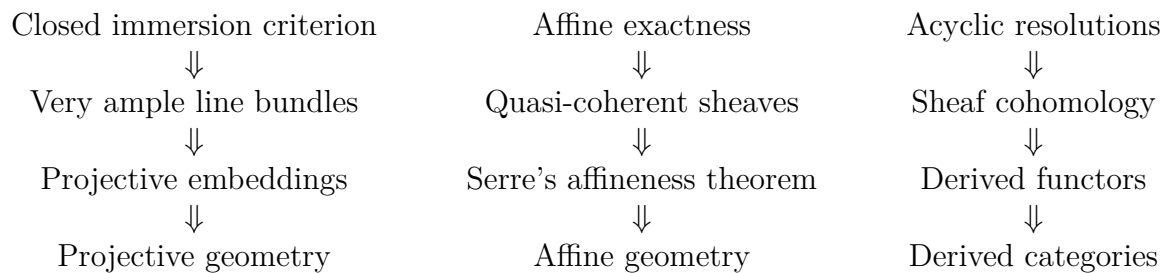
Serre duality,

Hodge theory,

and

modern derived algebraic geometry.

11.11 Global Concept Map



11.12 Final Summary

The three problems together form a structural triangle:

$$\boxed{\text{Projective geometry} \quad \text{Affine geometry} \quad \text{Cohomology}}$$

The first problem explains how sections give embeddings into projective space.

The second problem explains why affine schemes are controlled by modules.

The third problem explains how cohomology is computed using resolutions.

Together, they express one of the central principles of algebraic geometry:

$$\boxed{\text{global sections encode geometry, algebra, and cohomology.}}$$

12 Examples for the Extensions and Connections

12.1 Example: Very Ample Line Bundles

Consider

$$X = \mathbb{P}_k^1, \quad \mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1).$$

The global sections are

$$H^0(X, \mathcal{O}(1)) = \langle x_0, x_1 \rangle.$$

The associated morphism is

$$[x_0 : x_1] \mapsto [x_0 : x_1].$$

This is a closed immersion, so

$$\mathcal{O}_{\mathbb{P}^1}(1)$$

is very ample.

Similarly,

$$\mathcal{O}_{\mathbb{P}^n}(1)$$

is very ample on every projective space.

By contrast, on

$$X = \mathbb{P}^1,$$

the trivial bundle

$$\mathcal{O}_X$$

has only constant sections and cannot separate points. Therefore it is not very ample.

12.2 Example: Projective Embeddings from Graded Rings

Let

$$S = k[x, y].$$

Then

$$X = \text{Proj}(S) = \mathbb{P}_k^1.$$

The degree-one piece is

$$S_1 = \langle x, y \rangle.$$

These homogeneous coordinates define the standard embedding

$$\mathbb{P}^1 \hookrightarrow \mathbb{P}^1.$$

Using the degree-two piece

$$S_2 = \langle x^2, xy, y^2 \rangle,$$

one obtains the Veronese embedding

$$\mathbb{P}^1 \hookrightarrow \mathbb{P}^2.$$

Thus different graded pieces produce different projective realizations of the same variety.

12.3 Example: Tangent Spaces and Differentials

Consider the parabola

$$X = V(y - x^2) \subseteq \mathbb{A}^2.$$

At the origin

$$P = (0, 0),$$

the maximal ideal is

$$\mathfrak{m}_P = (x, y).$$

The cotangent space is

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

Since

$$y = x^2,$$

we have

$$y \in \mathfrak{m}_P^2.$$

Therefore

$$\mathfrak{m}_P / \mathfrak{m}_P^2 = \langle x \rangle.$$

Hence

$$\dim T_P X = 1.$$

The tangent line is simply the x -axis.

This illustrates how first-order information is extracted from

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

12.4 Example: Serre's Affineness Criterion

Let

$$X = \operatorname{Spec} k[x].$$

Then X is affine, and for every quasi-coherent sheaf $\mathcal{F}$,

$$H^i(X, \mathcal{F}) = 0 \quad (i > 0).$$

For instance,

$$H^1(X, \mathcal{O}_X) = 0.$$

Now compare with

$$X = \mathbb{P}^1.$$

Although

$$H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0,$$

one has

$$H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2)) \neq 0.$$

Hence $\mathbb{P}^1$ is not affine.

This demonstrates the difference between affine and projective geometry.

12.5 Example: Čech Cohomology

Cover

$$\mathbb{P}^1 = U_0 \cup U_1$$

where

$$U_0 = D_+(x_0), \quad U_1 = D_+(x_1).$$

For the sheaf

$$\mathcal{O}_{\mathbb{P}^1},$$

the Čech complex is

$$0 \rightarrow \mathcal{O}(U_0) \oplus \mathcal{O}(U_1) \rightarrow \mathcal{O}(U_0 \cap U_1) \rightarrow 0.$$

Explicitly,

$$0 \rightarrow k[t] \oplus k[t^{-1}] \rightarrow k[t, t^{-1}] \rightarrow 0.$$

The resulting cohomology is

$$H^0(\mathbb{P}^1, \mathcal{O}) = k,$$

$$H^1(\mathbb{P}^1, \mathcal{O}) = 0.$$

Thus Čech cohomology reproduces sheaf cohomology.

12.6 Example: Derived Functors

Consider the global section functor

$$\Gamma(X, -).$$

For an affine scheme,

$$R^i\Gamma(X, \mathcal{F}) = 0 \quad (i > 0).$$

However, on

$$X = \mathbb{P}^1,$$

the sheaf

$$\mathcal{O}_{\mathbb{P}^1}(-2)$$

satisfies

$$R^1\Gamma(\mathbb{P}^1, \mathcal{O}(-2)) = H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k.$$

Thus higher derived functors measure the failure of exactness of global sections.

12.7 Example: Injective Resolution

For a sheaf $\mathcal{F}$, choose an injective resolution

$$0 \rightarrow \mathcal{F} \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots .$$

For example, if

$$\mathcal{F} = \underline{\mathbb{Z}}$$

on a topological space, then

$$H^1(X, \underline{\mathbb{Z}})$$

can be computed from

$$0 \rightarrow \Gamma(I^0) \rightarrow \Gamma(I^1) \rightarrow \Gamma(I^2) \rightarrow \dots .$$

This is the standard definition of sheaf cohomology in homological algebra.

12.8 Example: Derived Categories

A sheaf can be regarded as a complex concentrated in degree 0:

$$\dots \rightarrow 0 \rightarrow \mathcal{F} \rightarrow 0 \rightarrow \dots .$$

For example,

$$\mathcal{O}_{\mathbb{P}^1}$$

may be replaced by an injective resolution

$$I^\bullet.$$

Inside the derived category,

$$\mathcal{O}_{\mathbb{P}^1} \simeq I^\bullet.$$

Both objects produce the same cohomology.

This is the basic idea behind derived categories: replacing objects by equivalent resolutions.

12.9 Example: Grassmannians from Vector Bundles

Consider

$$X = \mathbb{P}^1$$

and the rank-two bundle

$$\mathcal{E} = \mathcal{O}(1) \oplus \mathcal{O}(1).$$

Its global sections form a four-dimensional vector space.

These sections define a map

$$X \rightarrow \mathrm{Gr}(1, 3).$$

Instead of assigning a point in projective space, each point of X determines a line in a fixed vector space.

This generalizes the construction of projective embeddings from line bundles.

12.10 Example: Serre Duality

Let

$$X = \mathbb{P}^1.$$

The canonical bundle is

$$\omega_X = \mathcal{O}_{\mathbb{P}^1}(-2).$$

Serre duality states

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong H^0(\mathbb{P}^1, \mathcal{O})^\vee.$$

Since

$$H^0(\mathbb{P}^1, \mathcal{O}) = k,$$

we obtain

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k.$$

Thus a nontrivial first cohomology group is dual to a global section space.

This is one of the simplest manifestations of Serre duality.

12.11 Summary Table

Extension	Typical Example	Main Idea
Very ample bundle	$\mathcal{O}_{\mathbb{P}^1}(1)$	Embeds into projective space
Projective embedding	Veronese map	Use graded pieces
Tangent spaces	$y = x^2$	$\mathfrak{m}/\mathfrak{m}^2$
Affineness	$\operatorname{Spec} k[x]$	$H^{>0} = 0$
Čech cohomology	$\mathbb{P}^1 = U_0 \cup U_1$	Compute via overlaps
Derived functors	$H^1(\mathbb{P}^1, \mathcal{O}(-2))$	$R^i\Gamma$
Injective resolutions	$0 \rightarrow \mathcal{F} \rightarrow I^\bullet$	Define cohomology
Derived categories	$\mathcal{F} \simeq I^\bullet$	Equivalent complexes
Grassmannians	$\mathcal{O}(1) \oplus \mathcal{O}(1)$	Higher-rank analogue
Serre duality	$\mathbb{P}^1$	$H^1 \longleftrightarrow (H^0)^\vee$